What the cfRNA signals tell us about TB

An illustrated interpretation of the completed gene-interaction follow-up notebooks. Each analysis begins with its biological question and then explains the statistical details.

The strongest supported finding

Cell-free RNA (cfRNA) is RNA found outside cells, here measured in blood plasma. It combines material from different cell and tissue sources. **Tuberculosis (TB)** can be associated with changes in this mixture and in expression within its sources.

TB-positive participants show a broader **interferon-related RNA expression pattern**, extending beyond the six XGBoost-selected genes. The interferon-gamma pathway score remains higher after broad RNA-composition adjustment in training, test, and validation: approximately **0.95, 0.76, and 1.02 training standard deviations**.

At the same time, the SPI1–MNDA–LYZ signal closely tracks estimated myeloid RNA contribution. Stable patient groups nearly reproduce cohort membership. These observations support a mixture of disease-associated expression and changing RNA sources; they do not isolate a cellular mechanism.

The deconvolution data already exist

discovery_cell_types.tsv contains plasma cfRNA source estimates for 268 samples, including all 251 participants used here. The completed notebooks already use those estimates. **cto_wb.txt** contains a separate whole-blood RNA dataset for 60 validation participants.

Analysis decision: use the existing plasma estimates as the primary composition input. The BayesPrism gene-exclusion job is an **optional robustness check**; it does not replace those estimates or delay interpretation of the completed results.

How to navigate this report

Pages 2–4: participants, input data, measurement units, and the existing deconvolution files.

Pages 5–10: what XGBoost uses, how the three-gene signal relates to RNA sources, and the initial adjustment models.

Pages 11–14: heatmap, PCA, UMAP, and patient clustering.

Pages 15–18: cohort effects, finer myeloid adjustment, and Xpert Ultra burden checks.

Pages 19–22: pathway methods, enrichment, and participant-level pathway scores.

Pages 23–27: the optional sensitivity analysis, biological synthesis, glossary, pathway labels, and sources.

Scope and evidence status

This report includes all **17 completed figures** from the four biological follow-up notebooks, plus the methods and current status of the deconvolution notebook. It does not rerun model selection or claim new deconvolution results.

The names “test” and “validation” preserve the original split labels. Both datasets have already been inspected during this research, so these biological follow-ups are **exploratory**, rather than fresh confirmatory validation.

Which participants and measurements are being compared?

A sample is one participant’s plasma RNA profile. A split is an analysis role; a cohort is a study population or collection context.

Split	Cohort	All participants	TB positive	TB negative	Role in these analyses
Training	GHL	63	39	24	Model fitting and training-derived transformations
Training	R2D2	67	34	33	Same training split; a second cohort
Test	GHL	30	18	12	Apply the frozen model or transformation
Test	R2D2	31	16	15	Apply the frozen model or transformation
Validation	TBSQUARE	60	35	25	A different cohort; evaluated separately
Total	Three cohort labels	251	142	109	130 training + 61 test + 60 validation

What each input contributes

counts.train/test/val.tsv: gene-by-sample sequencing read counts. A read is a sequenced RNA fragment assigned to a gene. Counts reflect both RNA abundance and sequencing depth; they are not directly comparable across samples without normalization.

metadata.train/test/val.tsv: participant identifiers, TB labels, cohort, and available clinical variables. TB negative is the study’s comparison category; it should not automatically be read as “healthy.”

discovery_cell_types.tsv: estimated relative contributions from 41 reference RNA sources. The count and fraction tables describe the same plasma assay from different perspectives.

Why the split/cohort distinction matters

Training and test each contain GHL and R2D2; validation contains only TBSQUARE. Therefore, a difference between validation and the other splits can reflect cohort composition as well as sampling variation. There is no within-validation contrast between cohorts.

All expression and fraction joins use sample identifiers. Age, sex, and HIV are included where stated. Sputum Xpert Ultra categories are available for **135 of the 142 TB-positive participants**; those analyses have smaller denominators.

No classifier is retrained for the clinical or pathway association tests. Their outcome is an expression score, and TB status is a predictor. That direction differs from a classifier predicting TB from expression.

Learn the units before comparing the numbers

Different figures quantify different things. An expression score, a prediction contribution, a correlation, and an adjusted difference are not interchangeable.

Expression and standardization

VST means variance-stabilizing transformation. DESeq2 normalizes counts and transforms them so the variability is less dependent on mean abundance. The resulting values are expression units, not RNA molecule counts or probabilities.

A **training z score** subtracts a gene's training mean and divides by its training standard deviation (SD): z = (expression − training mean) / training SD. A value of +1 is one training SD above that gene's training mean.

The **three-gene signal** is $(z_{SPI1} + z_{MNDA} + z_{LYZ}) / 3$, with fixed equal positive weights. A one-unit difference is one mean gene-level z-score unit. The composite itself has not been rescaled to SD = 1.

A **pathway score** averages training-standardized genes in a pathway, then divides by the SD of that average in training. Here a one-unit difference really is one training SD of the pathway summary.

Predictions and associations

Log odds are log[p / (1 − p)], where p is predicted TB probability. Here prediction log odds = model baseline + sum of SHAP contributions. A SHAP value of +1 is not a one-unit probability increase; the probability change depends on the starting probability.

An **adjusted TB coefficient** estimates the TB-positive minus TB-negative difference in an expression score at the same modeled covariate values. It is not a classifier AUC and is not a causal effect of TB.

Mixtures, correlations, and uncertainty

A **RNA-source fraction** is a relative contribution to the modeled RNA mixture. A fraction of 0.30 means 30% of that modeled contribution, not 30% of circulating cells. Fractions sum to one, so one source's increase changes the relative share of others.

ALR, an additive log ratio, represents composition as log(source group / reference group). The primary models use myeloid, lymphoid, and erythroid/platelet ratios, all relative to the remaining sources. A one-unit change in a natural-log ratio multiplies that ratio by e ≈ 2.72.

Spearman ρ compares ranks: do participants with higher values of one measurement tend to have higher values of another? It ranges from −1 to +1. A partial rank correlation first removes modeled cohort and TB-status differences from both sets of ranks.

A **95% confidence interval** reports uncertainty in an estimate under the model assumptions. Intervals crossing zero are compatible with either direction or no difference at this precision; this does not prove no association.

HC3 is a regression uncertainty calculation that allows residual variability to differ across participants and adjusts for influential observations. Intervals here use a t distribution with the model's residual degrees of freedom.

A **p value** measures incompatibility with a specified null model, not the probability that a claim is false. **False discovery rate (FDR)** is the expected proportion of false positives among declared findings, under the procedure's assumptions. q < 0.05 is not a 5% probability that an individual claim is false. The correction family must be stated.

Yes: the manuscript folder already contains deconvolution datasets

The completed biological analyses reuse the plasma fractions. The current rerun is an additional sensitivity check, not the first estimate of cell-type contributions.

Existing file in Files_For_Manuscript	What it contains	Coverage	How it is used
discovery_cell_types.tsv	Plasma cfRNA: 41 estimated source fractions plus sample_id	268 rows; all 251 analysis samples present	Primary existing composition input to the completed follow-ups
cto_wb.txt	Whole-blood RNA: the same 41 source labels plus sample_id	60 rows; IDs map to all 60 TBSQUARE validation participants	A distinct assay; not a duplicate of plasma fractions
counts.train.tsv / counts.test.tsv / counts.val.tsv	Raw gene read-count matrices	130 / 61 / 60 participants	Mixture inputs, not deconvolution outputs
gene_interactions_reference/deconvolution_reference.rds	Prepared reference expression profiles and reference information	41 collapsed source types	Input cache for the fresh sensitivity analysis; not patient fractions

What was checked

Both original fraction tables have **42 columns**: an identifier and 41 numeric fractions whose row totals are approximately one. All analysis samples have plasma estimates. The 17 additional plasma rows are outside the current train/test/validation sample set.

The whole-blood identifiers were normalized to match TBSQUARE participants. Across the 60 matched rows × 41 fractions, **0 of 2,460 values are exactly identical** to plasma. Figure_3.ipynb also loads and labels these as separate plasma and WBRNA inputs.

Hidden checkpoint copies of these files are byte-identical backups. They do not add samples or independent measurements. Derived grouping tables in the figure outputs are summaries of the original plasma data.

Why an additional run can still be useful

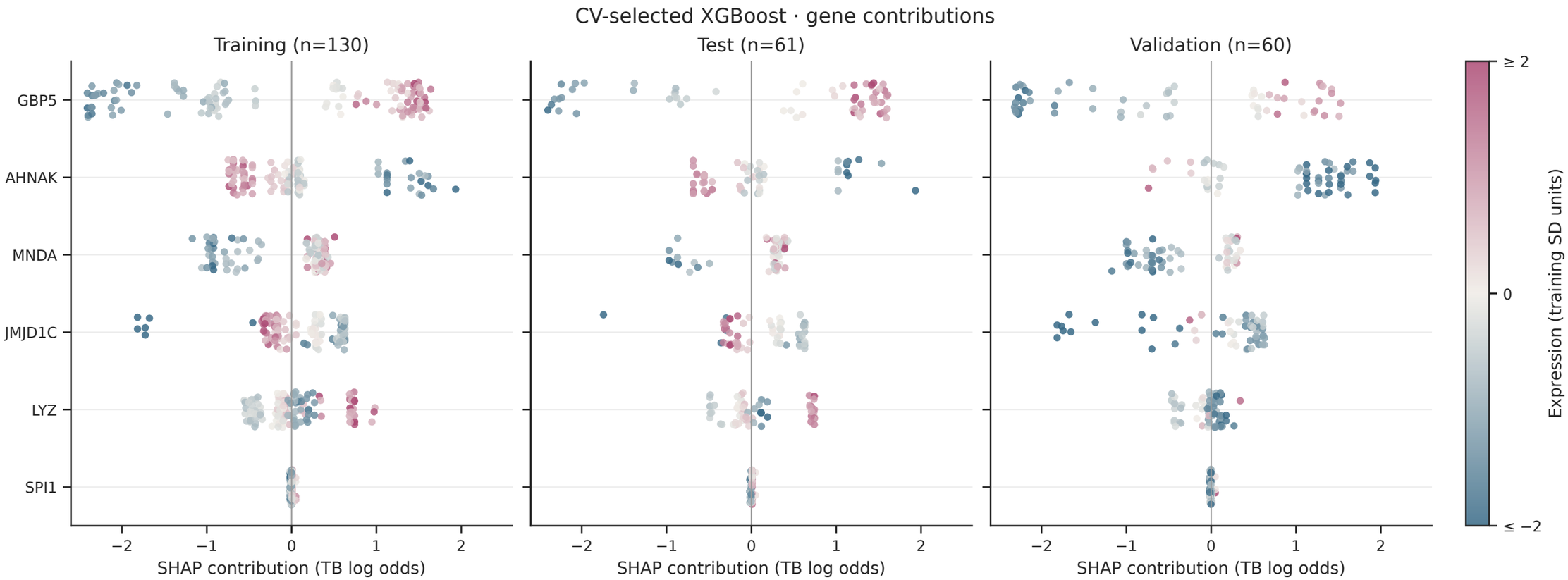
An existing fraction table does not record what the estimates would have been if GBP5, AHNAK, JMJD1C, MNDA, LYZ, and SPI1 had been absent from the input. To test that, the notebook fits a matched baseline and a six-gene-excluded version for each participant.

The fresh baseline is needed because software version, gene-ID harmonization, and the exact historical filtering branch cannot all be assumed identical. Comparing the original table directly with a newly excluded fit would confound those changes with gene removal.

The existing plasma data are the primary composition input. The additional gene-exclusion run is optional and does not gate these analyses. Whole-blood estimates remain a separate assay for paired comparisons, not a substitute plasma input. Neither dataset is a measurement of cell counts.

Which genes most influence the fitted XGBoost predictions?

Figure 1. SHAP (SHapley Additive exPlanations) decomposes each participant’s model prediction into contributions from the six selected genes.



Read the figure

The three panels contain training (130), test (61), and validation (60) participants. Each dot is one participant for one gene. Moving right means a larger contribution toward a TB-positive prediction; left means a contribution away from TB.

Color shows that gene’s training-standardized expression, with shared color limits. Vertical stacking separates overlapping dots; it is not an additional biological measurement. Genes are ordered by their mean absolute training SHAP value.

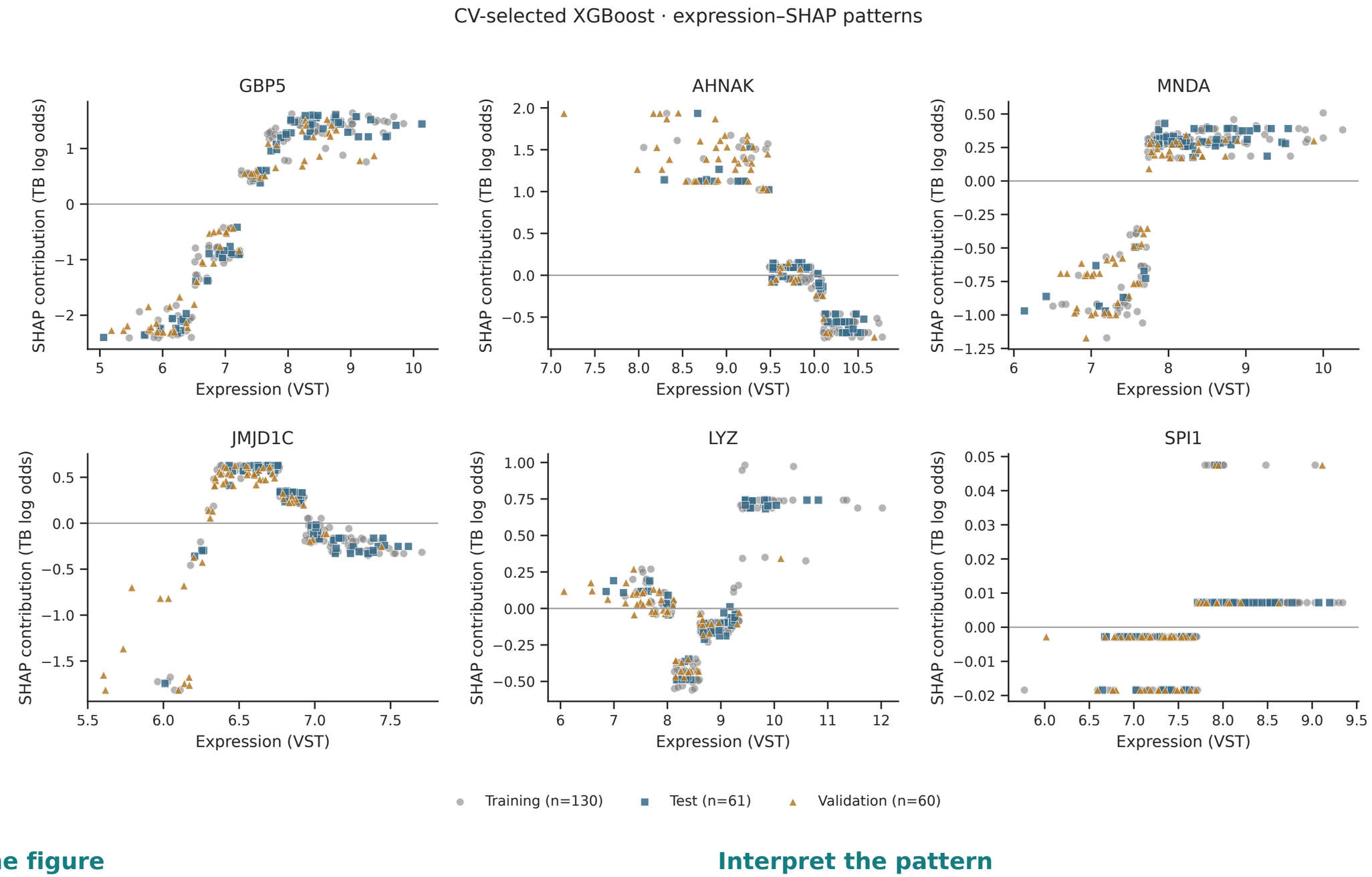
Interpret the pattern

GBP5 dominates the fitted prediction: mean absolute SHAP is 1.33, 1.36, and 1.30 log-odds units across the three splits. AHNAK has more influence in validation (1.03) than training (0.49). SPI1 contributes only about 0.01 in all splits.

These are contributions from the same frozen model, not separate models fitted to each split. A biologically interesting gene can have little incremental use in a particular fitted model. Large SHAP magnitude measures model influence, not causal importance or predictive accuracy. [2]

How does expression relate to each gene’s contribution?

Figure 2. Dependence plots show the observed relationship between expression and SHAP attribution for each gene.



Read the figure

Each panel is one gene. The horizontal axis is VST expression; the vertical axis is its SHAP contribution in TB log odds. Points retain their split colors and markers. Axis limits differ among genes, so compare the numerical scales, not only the height of the cloud.

Steps and plateaus are expected from decision trees. Vertical spread at similar expression can arise because other features affect how the model assigns a contribution.

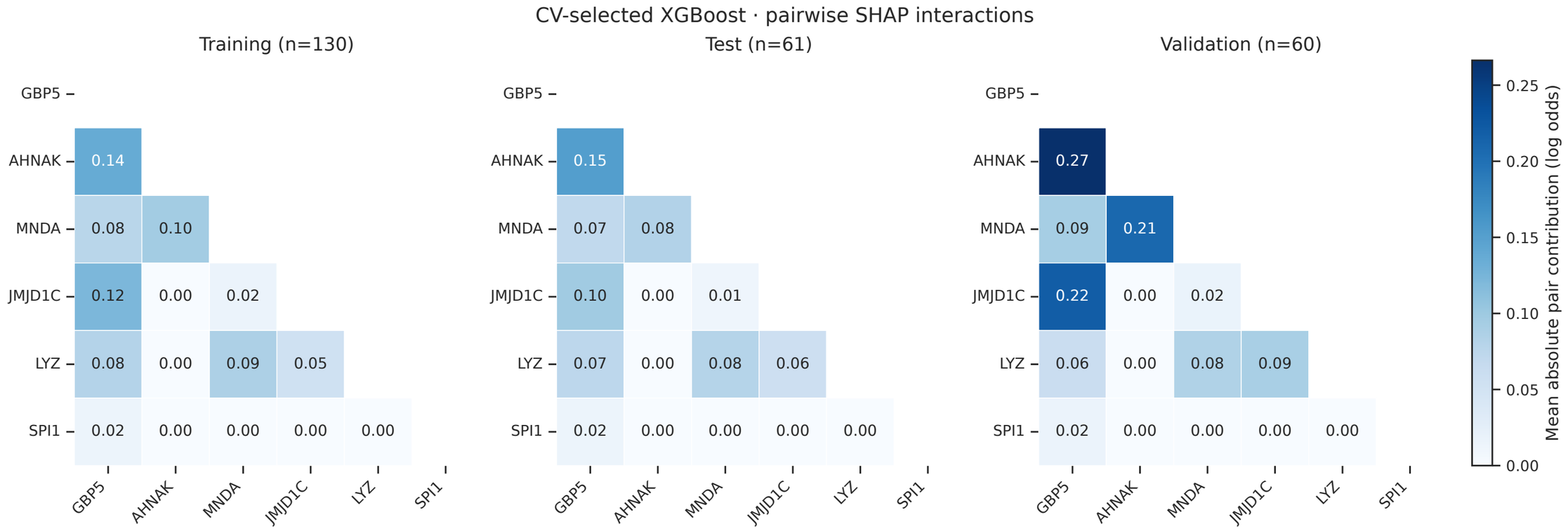
Interpret the pattern

Higher GBP5 and, broadly, MNDA expression tend to support TB predictions. Lower AHNAK supports TB. JMJD1C is most positive in an intermediate range, and LYZ is nonmonotonic. SPI1’s visible fluctuations are much smaller on its numerical scale.

These curves explain the observed fitted model. They do not show what would happen biologically if expression were experimentally changed, and tree thresholds are not established biological cutoffs.

Which gene pairs contribute nonadditively to the predictions?

Figure 3. Pairwise TreeSHAP interactions describe departures from additive feature contributions within this XGBoost model.



Read the figure

Each cell represents a pair of genes. The number and color give the mean absolute pair contribution across the participants in that split. Only one triangle is shown because the matrix is symmetric; the diagonal main effects are excluded.

For an off-diagonal TreeSHAP entry I_{ij}, the displayed pair magnitude is mean($|2I_{ij}|$). The factor of two combines the symmetric contributions. Taking an absolute value removes the direction of the interaction.

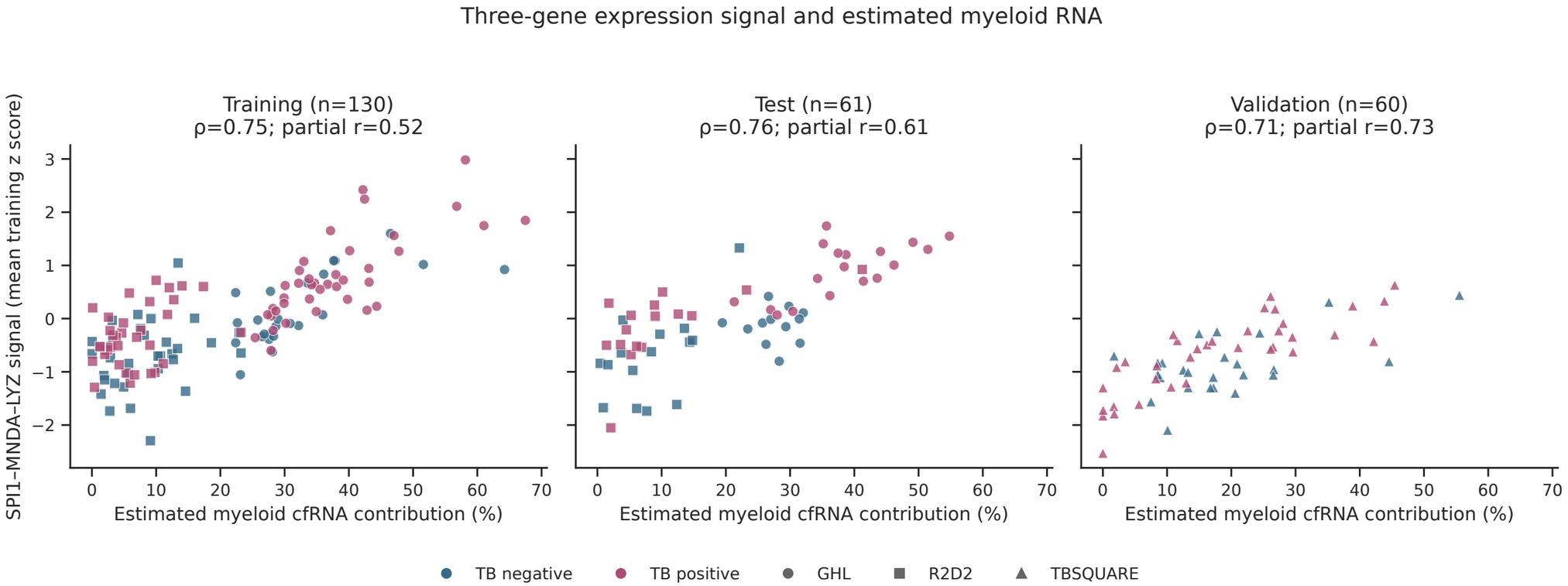
Interpret the pattern

GBP5–AHNAK is the strongest average pair: approximately 0.14, 0.15, and 0.27 log-odds units in training, test, and validation. Several other GBP5 pairs contribute; SPI1-related interactions are very small.

A model interaction is not evidence of a physical interaction or genetic epistasis. Correlated expression, thresholds, and cohort distributions can all contribute. A displayed 0.00 may reflect rounding rather than exact absence. [2]

Does the three-gene signal track myeloid RNA contribution?

Figure 4. The fixed SPI1–MNDA–LYZ score is compared with existing plasma deconvolution estimates.



Read the figure

Each dot is one participant. The horizontal axis is the estimated myeloid fraction multiplied by 100; the vertical axis is the mean training z score of SPI1, MNDA, and LYZ. Colors show TB status; shapes show cohort.

Myeloid combines seven reference labels: monocyte, macrophage, neutrophil, conventional dendritic cell, myeloid progenitor, basophil, and mast cell. “Myeloid” here is an operational sum of these reference contributions.

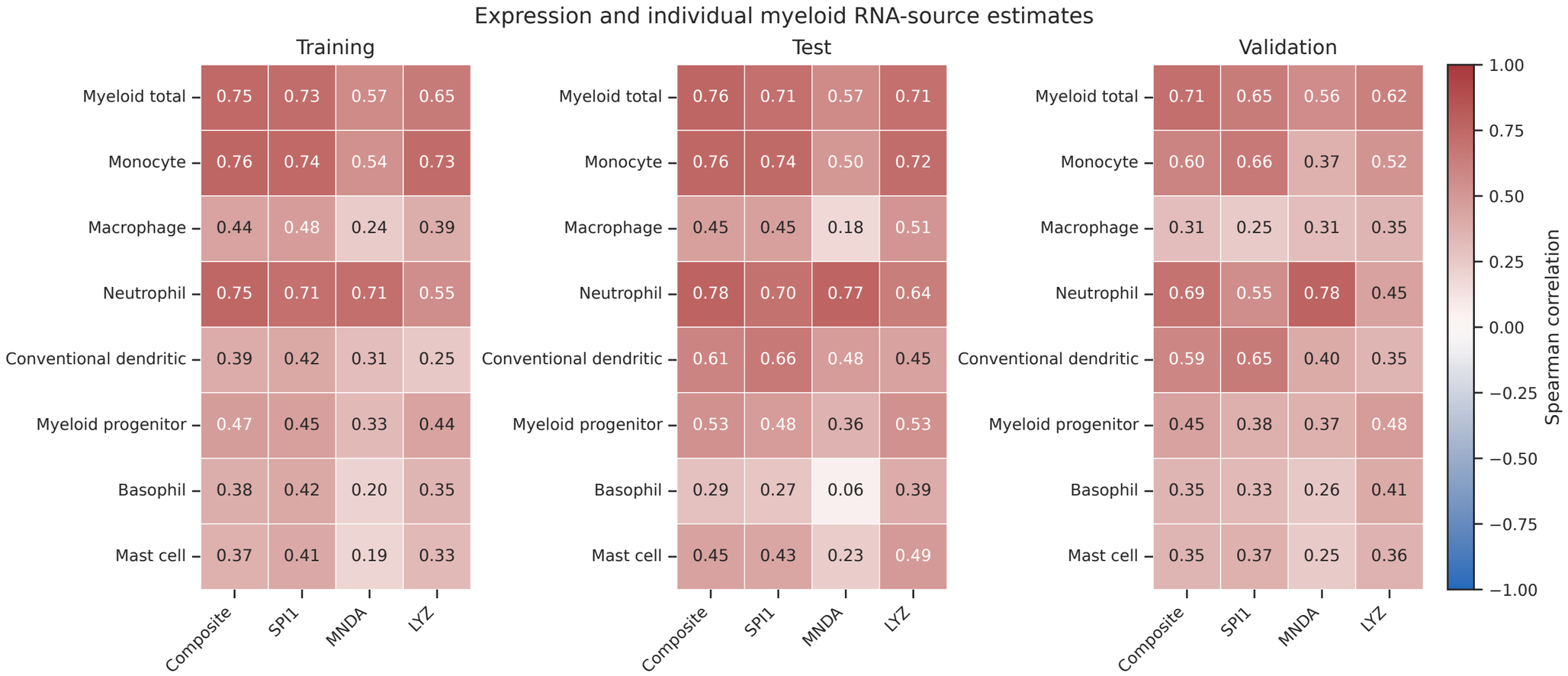
Interpret the pattern

Raw Spearman correlations are **0.75, 0.76, and 0.71** for training, test, and validation. Partial rank correlations after cohort/TB adjustment are **0.52, 0.61, and 0.73**. The association is not solely the result of separating TB labels or cohorts.

This is strong evidence that the score follows an estimated RNA-source mixture. However, expression and source estimates derive from the same RNA assay. Shared marker genes and correlated expression can contribute; this is not an independent count of myeloid cells.

Which estimated myeloid sources track the signal?

Figure 5. Source-specific rank correlations help describe which parts of the estimated mixture align with the score.



Read the figure

Rows show the total myeloid contribution and its seven component sources. Columns show the composite score and each of its genes. Each number is a Spearman rank correlation within one split; the color scale is shared from −1 to +1.

These are raw correlations, not partial correlations, regression coefficients, or percentages. The composite column is not the XGBoost prediction. No significance threshold is encoded in this heatmap.

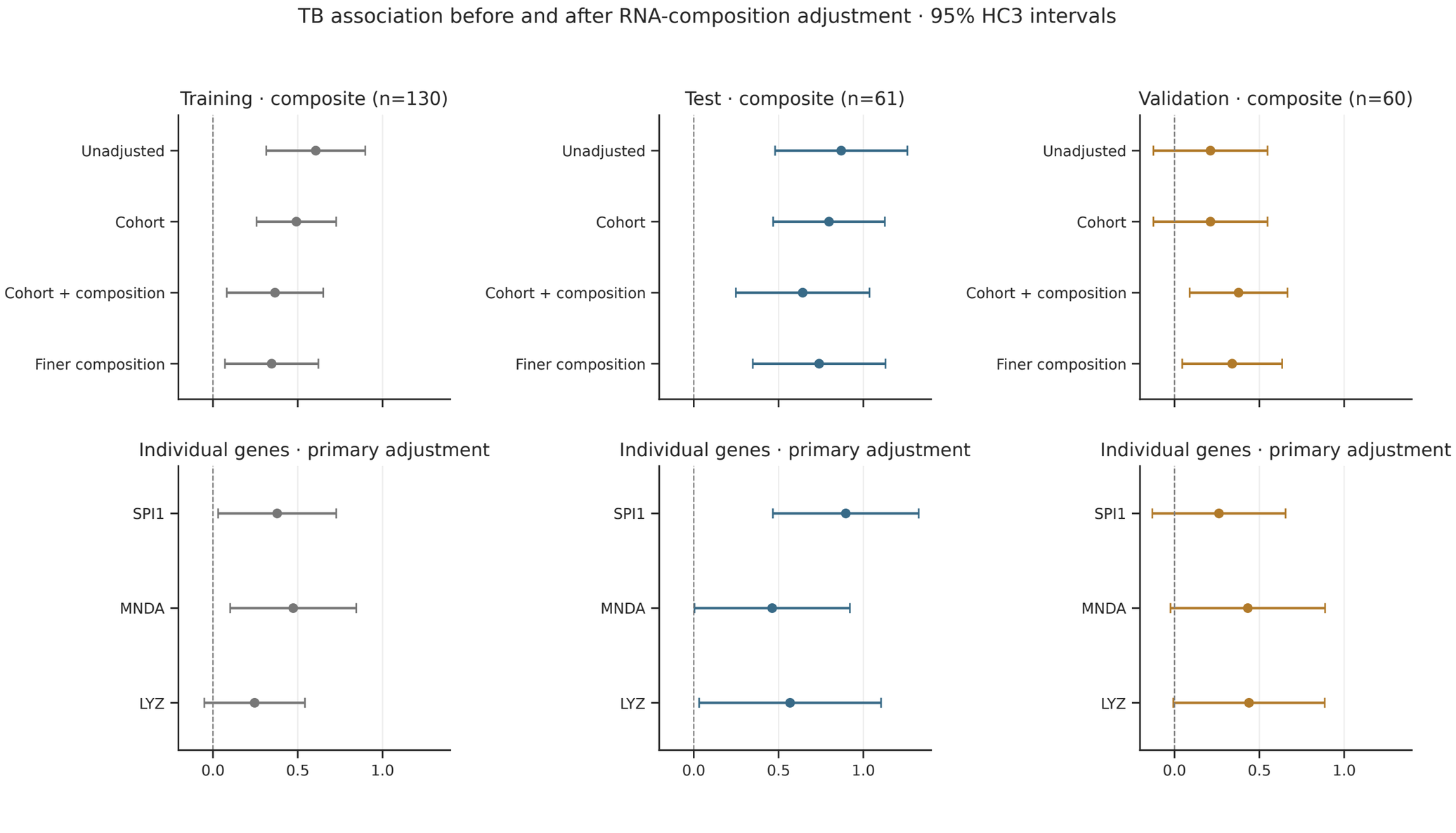
Interpret the pattern

Monocyte and neutrophil estimates show the clearest associations with the composite. Composite–monocyte correlations are approximately 0.76, 0.76, and 0.60; composite–neutrophil correlations are 0.75, 0.78, and 0.69.

This motivates distinguishing myeloid subgroups in later models. It does not identify a unique source: reference profiles share genes, source fractions are correlated, and all fractions are constrained by the same total mixture.

Does an adjusted TB association remain in the initial analysis?

Figure 6. The original regression comparisons ask how the TB-associated expression difference changes with cohort and composition adjustment.



Read the figure

Top panels compare the composite under four adjustment sets. Bottom panels show individual genes under the primary cohort-plus-composition model. Dots are TB-positive minus TB-negative differences; horizontal lines are 95% HC3 intervals. Zero means no modeled difference.

Composite differences use mean gene z-score units; individual-gene differences use that gene's training SD units. The primary composite estimates are **0.366, 0.642, and 0.378** across the three splits.

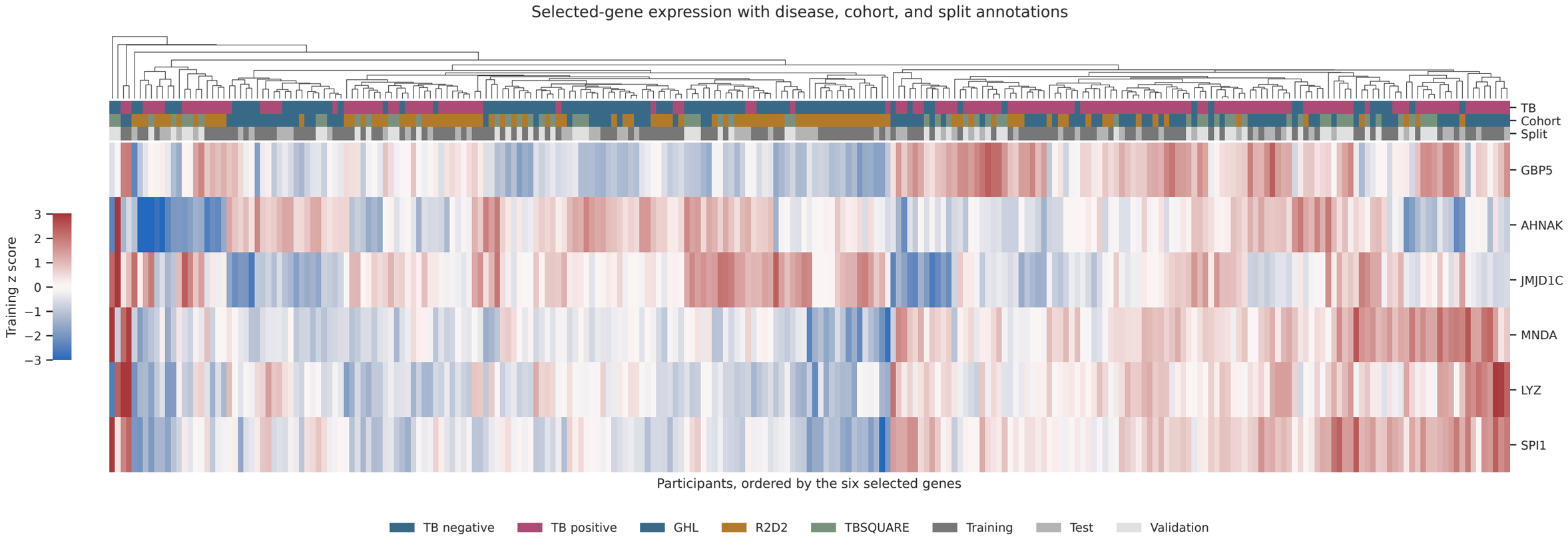
Interpret the pattern

The initial broad-composition model retains a positive composite association in each split. Validation's unadjusted interval crosses zero, whereas its adjusted interval does not. Adjustment can strengthen or weaken a conditional association.

"Finer composition" in this figure still pools all myeloid sources. It separates erythroid, platelet, and endothelial sources. The later clinical analysis instead separates monocyte and neutrophil contributions and gives a more cautious result. Individual validation-gene intervals here all cross zero.

How do the six selected genes vary across participants?

Figure 7. A heatmap places the expression panel alongside disease status, cohort, and split labels.



Read the figure

Rows are the six XGBoost-selected genes; each narrow column is a participant. Colors represent expression relative to each gene’s training mean and SD, with the displayed scale limited to −3 to +3. The annotation strips identify TB status, cohort, and split.

Participants are ordered by similarity across these six genes using all 251 samples for this descriptive display. The ordering itself does not define validated groups or a classifier.

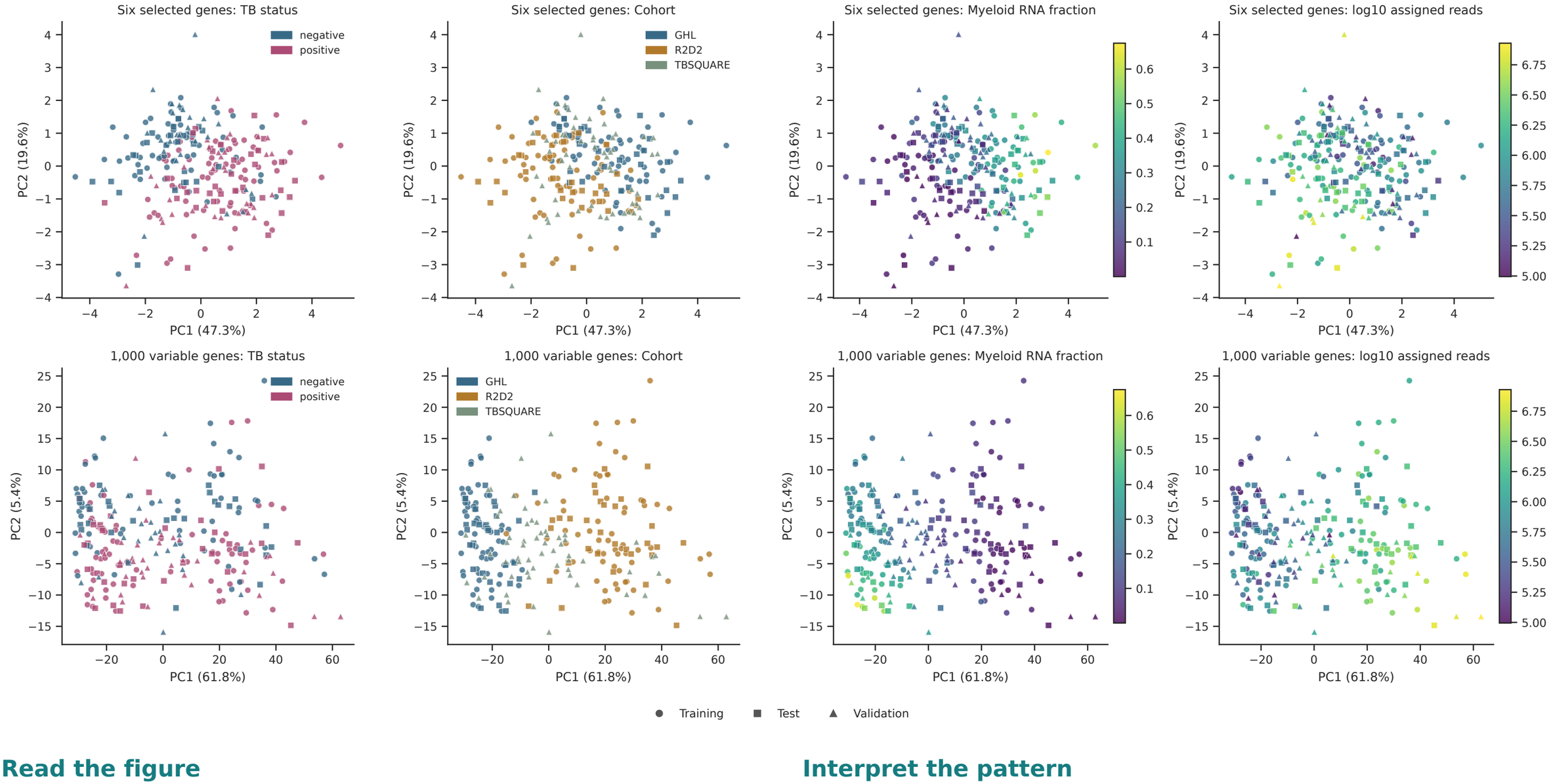
Interpret the pattern

The panel shows structured, heterogeneous expression rather than a single uniform increase across all six genes. Comparing expression blocks with the cohort strip helps reveal whether disease-related patterns overlap with population structure.

Because the genes were selected using training TB labels, visible TB separation is not independent evidence of their predictive value. This heatmap uses the training-reference VST, whereas the original XGBoost/SHAP analysis retains the original split-specific VST.

Which major expression patterns follow disease, cohort, or RNA sources?

Figure 8. Principal component analysis (PCA) compresses many correlated measurements into a few linear directions of variation.



Read the figure

The top row uses six selected genes; the bottom row uses the 1,000 most variable eligible genes in training. Each row repeats the same coordinates four times, colored by TB, cohort, myeloid fraction, and log10 assigned reads. Shapes identify splits.

PC1 and PC2 are weighted expression combinations fitted in training. For the broad panel they explain **61.8% and 5.4%** of standardized training expression variance. Those percentages are not prediction accuracy.

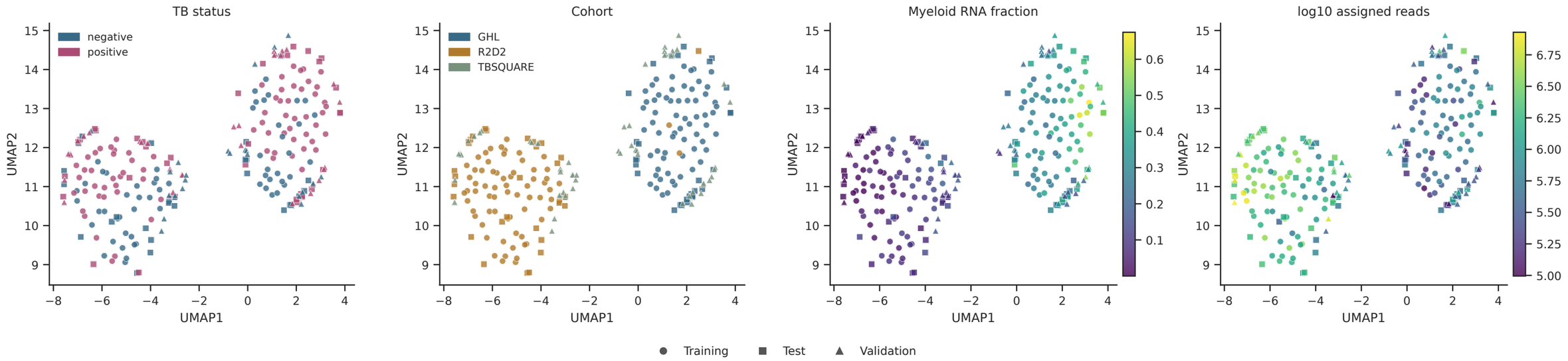
Interpret the pattern

The broad representation aligns strongly with cohort and estimated myeloid contribution; read depth also varies across the structure. Disease is not the only organizing feature. This does not prove the pattern is purely a technical batch effect.

Assigned reads are gene-assigned sequencing counts; a change of one on their log10 scale is a tenfold depth difference. Scaling and PCA are training-fitted, then applied to holdouts. Broad genes were selected by variability without using TB labels.

Does a nonlinear map reveal additional patient structure?

Figure 9. UMAP emphasizes local similarity in the 20-component broad expression representation.



Read the figure

UMAP means Uniform Manifold Approximation and Projection. The four panels show the same coordinates with different overlays: TB status, cohort, myeloid fraction, and log10 assigned reads. Each dot is a participant; shapes identify the split. UMAP axes have no direct biological units.

The map was fitted to training using 15 neighbors and minimum distance 0.3, then test and validation were transformed into that map. Checks also varied the neighborhood size to 10/30 and the random seed.

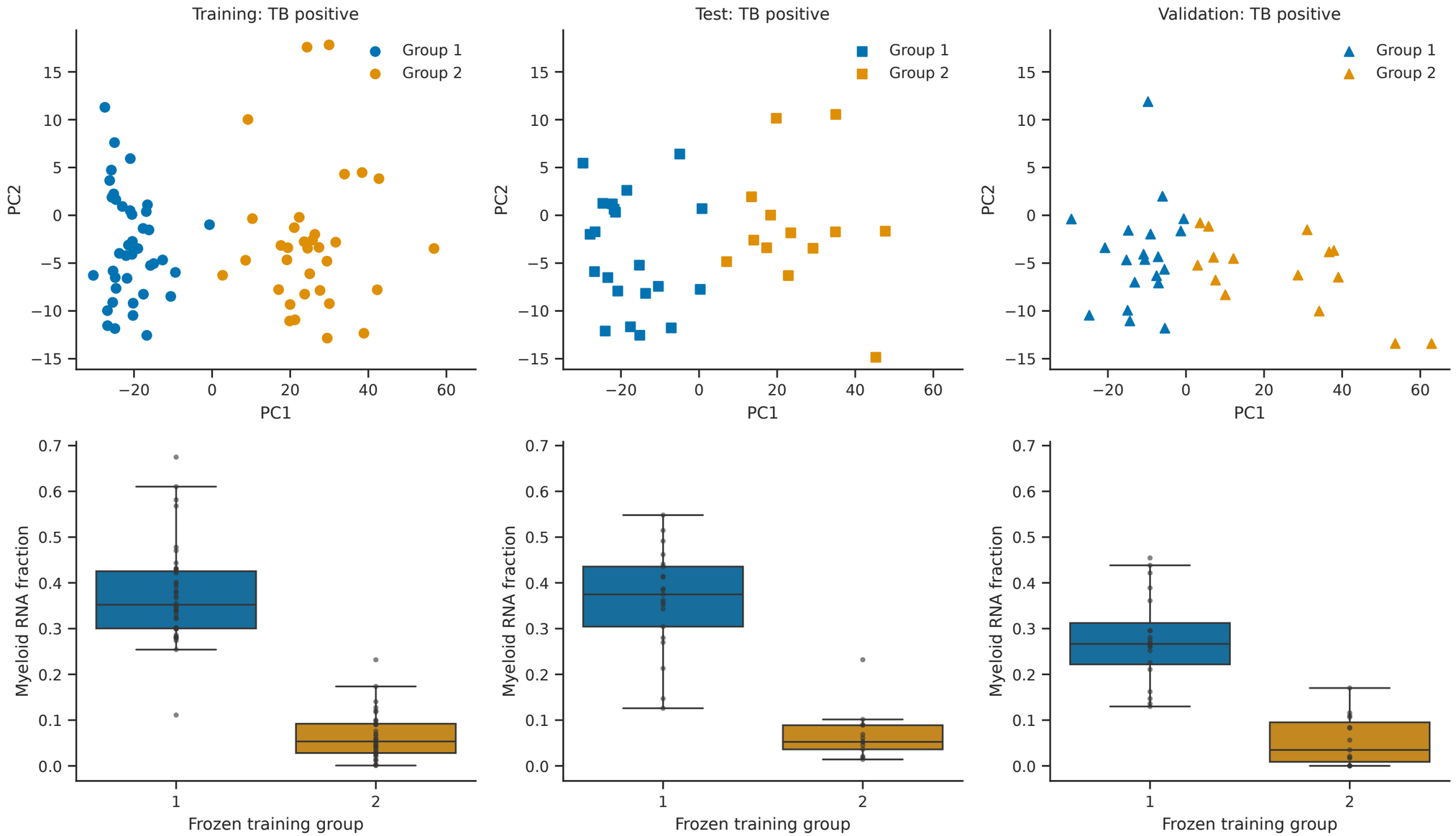
Interpret the pattern

The map repeats much of the cohort/composition structure seen with PCA. It provides an accessible display, not evidence that apparent islands are distinct TB subtypes. UMAP can distort density and introduce apparent gaps. [3]

Neighborhood trustworthiness is approximately **0.943, 0.941, and 0.857** by split. This measures how faithfully nearby points are represented, not TB accuracy. Patient clustering was performed in the 20-PC space, not by drawing boundaries around UMAP islands.

Are the stable TB-positive patient groups biological subtypes?

Figure 10. K-means groups were learned from the 73 TB-positive training participants and then held fixed.



Read the figure

Top panels display the group assignments in the first two broad PCs. Bottom panels compare estimated myeloid fractions between groups. Only TB-positive participants are shown: 73 training, 34 test, and 35 validation. Test/validation are assigned to the nearest frozen training center.

Among k = 2, 3, and 4, two groups had the highest training silhouette (0.547). Silhouette measures compactness/separation. Bootstrap ARI measures agreement of assignments after resampling participants; its median was 1.00.

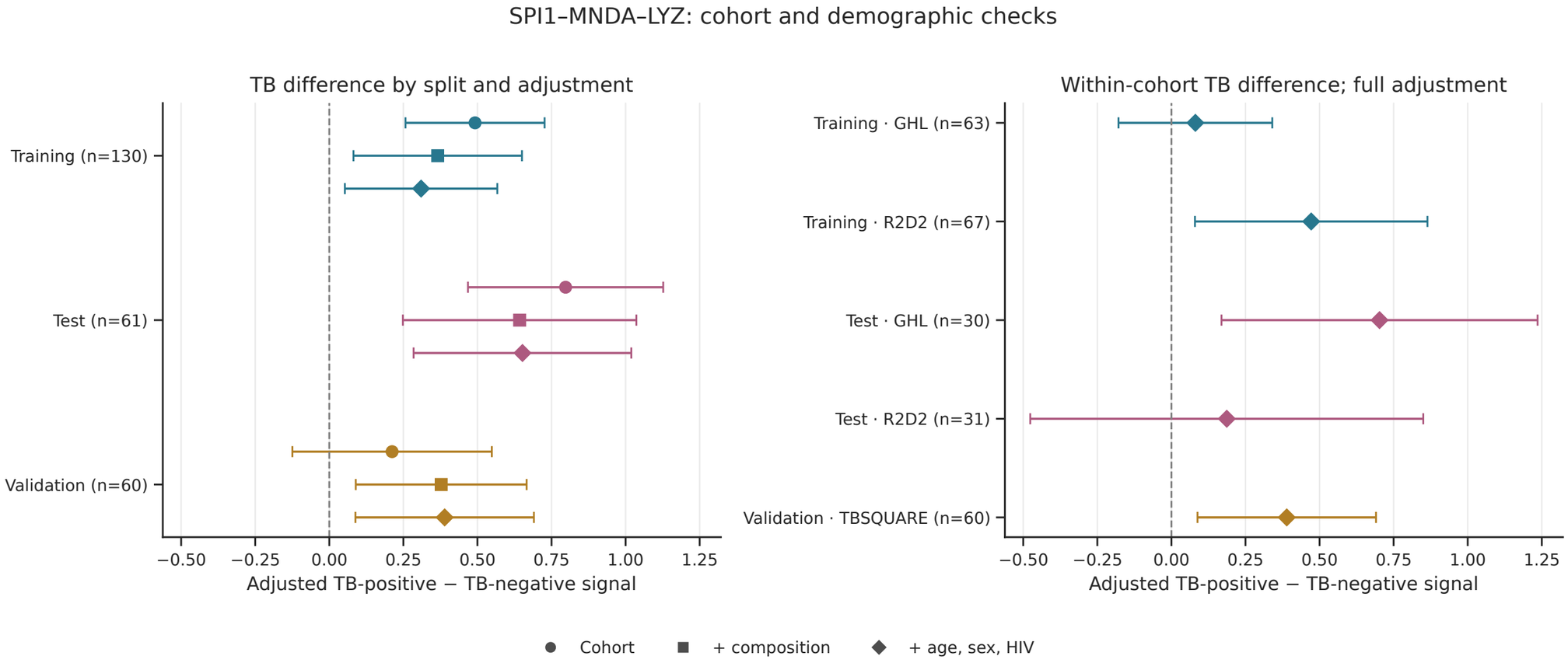
Interpret the pattern

The training groups almost reproduce cohort. Group 1 contains 39 GHL + 1 R2D2 participants; group 2 contains 33 R2D2. Cohort ARI is 0.945. Their median estimated myeloid contributions are 35.2% versus 5.3%.

Stability therefore does not establish disease subtypes. The bootstrap holds gene selection, scaling, and PCA fixed, so it measures conditional clustering stability. Assigning validation participants to a training center does not independently validate that center as a biological subtype.

How robust is the TB association across covariates and cohorts?

Figure 11. The clinical follow-up extends the original adjustment models with age, sex, HIV, and within-cohort comparisons.



Read the figure

The left panel compares adjustment sets within each split. The right panel fits the fully adjusted model separately within each cohort/split. Estimates are TB-positive minus TB-negative differences in the fixed three-gene score; intervals are 95% HC3 t intervals.

With broad composition plus demographics, estimates are **0.310 [0.053, 0.567]** in training, **0.652 [0.285, 1.019]** in test, and **0.389 [0.088, 0.691]** in validation.

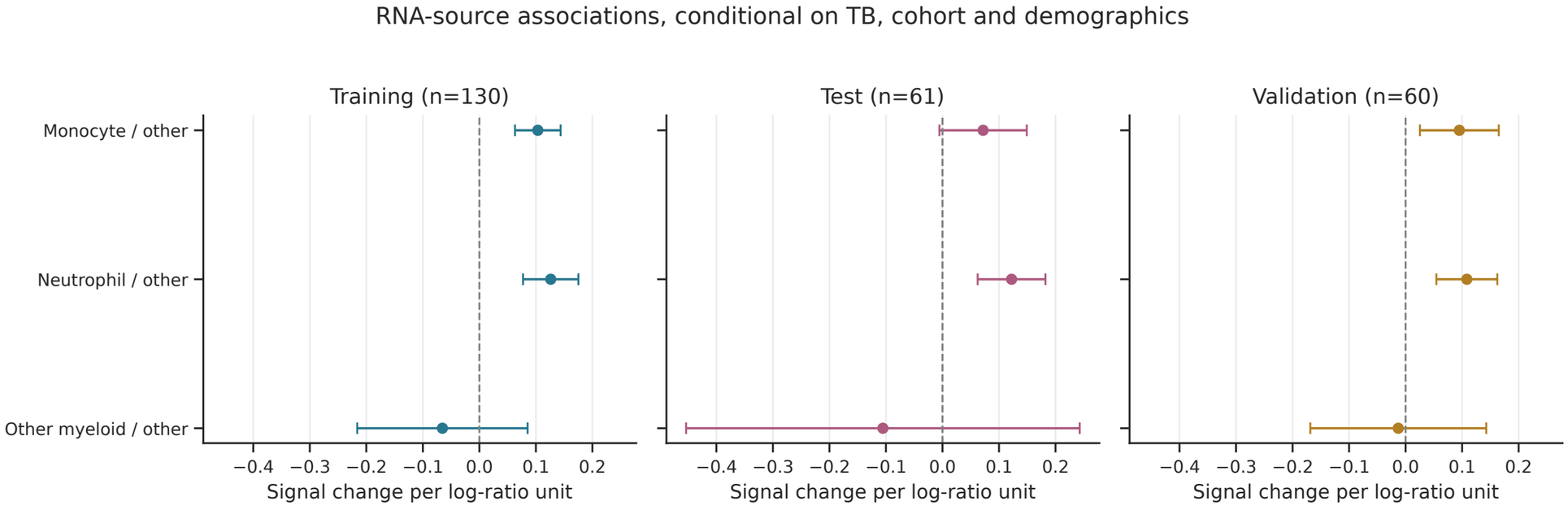
Interpret the pattern

All split-level intervals are positive, but training GHL and test R2D2 have within-cohort intervals that cross zero. The estimated R2D2-minus-GHL contrast changes direction between training and test; there is no stable replicated cohort interaction.

A separate read-depth sensitivity uses **log10 exonic reads**, not the assigned-read measure in the PCA plots. It changes the training estimate to 0.267 [−0.012, 0.546], while test/validation remain positive. These conditional associations depend on the adjustment specification.

What changes when monocytes and neutrophils are distinguished?

Figure 12. Separate myeloid source ratios test whether a pooled myeloid total leaves important mixture structure unaccounted for.



Read the figure

The plotted coefficients describe associations of the expression score with monocyte, neutrophil, and other-myeloid log ratios, conditional on TB, cohort, other source ratios, age, sex, and HIV. They are not effects per extra cell.

Monocyte and neutrophil coefficients are generally positive; the test monocyte interval crosses zero. All ratios use the same remaining-source denominator. One natural-log unit means a 2.72-fold ratio change at fixed other modeled ratios.

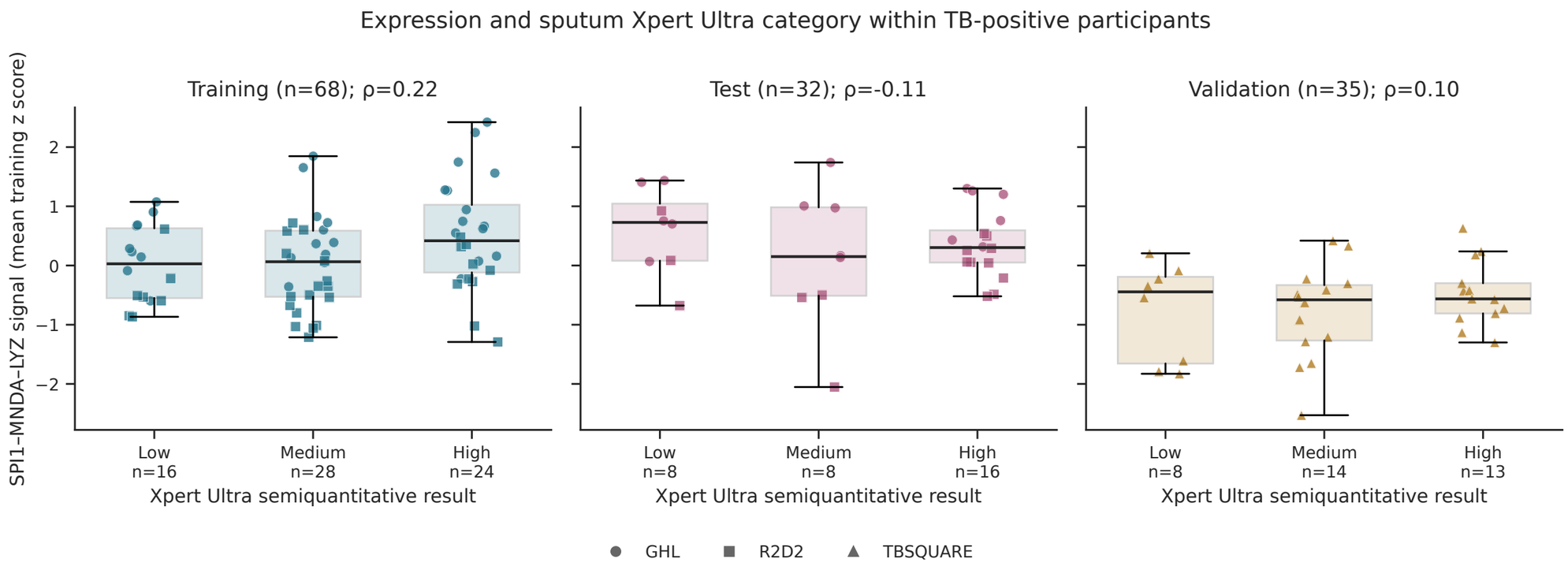
The key accompanying TB result

When these finer source ratios replace total myeloid, the TB coefficients become **0.170 [−0.031, 0.370]** in training, **0.536 [0.236, 0.836]** in test, and **0.176 [−0.129, 0.480]** in validation.

Training and validation are now compatible with no residual difference at this precision. This weakens the claim of composition-independent activation. It does not show that composition explains everything: correlated, estimated covariates and additional model parameters can affect both estimates and uncertainty.

Does the signal vary with sputum Xpert Ultra category?

Figure 13. This analysis asks about microbial-burden categories only among participants already classified as TB positive.



Read the figure

The horizontal axis is the recorded Xpert Ultra semiquantitative category: Low, Medium, or High. The vertical axis is the three-gene expression score. Each point is a participant, with cohort retained in the display.

Boxes show the middle 50% of values and the median; whiskers extend to observations within 1.5 interquartile ranges. Available denominators are **68 training, 32 test, and 35 validation**. Seven TB-positive participants lack results, all in GHL.

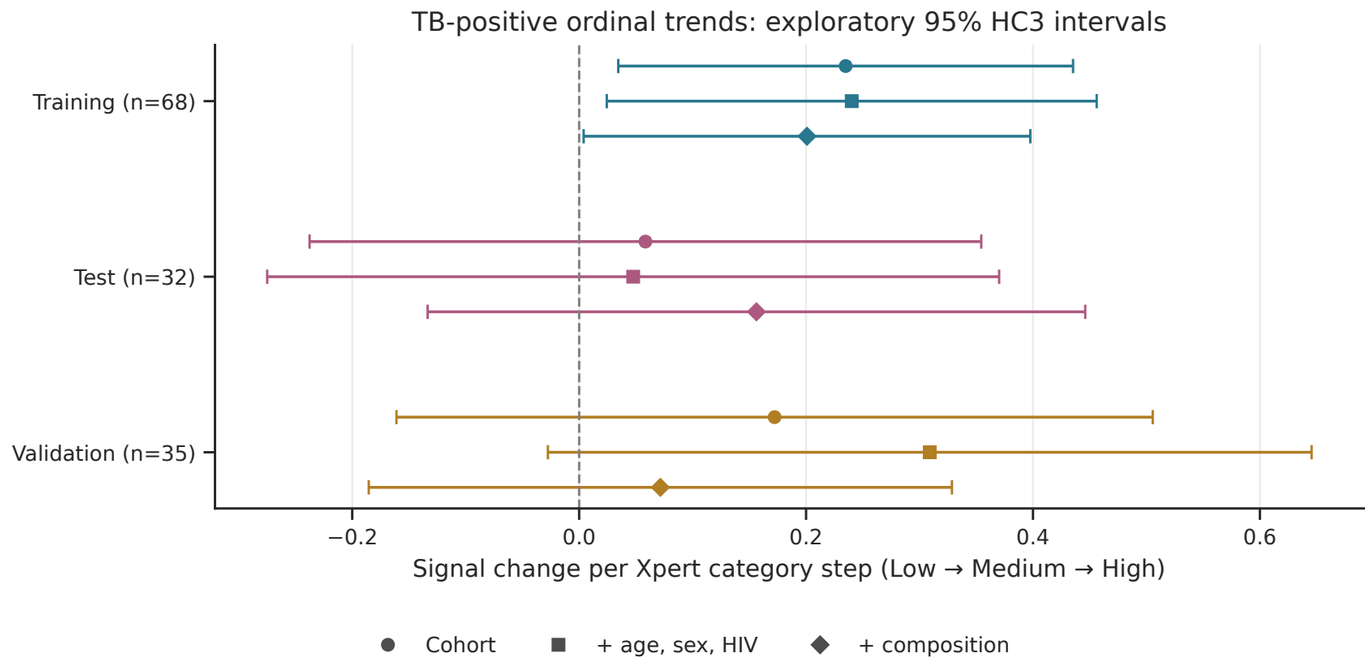
Interpret the pattern

There is no consistent increasing pattern across all three splits. Raw rank correlations with ordered category are about **0.22, −0.11, and 0.10**. The direction and size are not reproducible.

Xpert is a sputum microbial-burden proxy, not a count of circulating RNA sources or a complete clinical severity score. TB-negative participants are not assigned “zero burden,” and missing categories remain missing. Complete-case analysis cannot eliminate bias from missing measurements. [1]

Is there an adjusted trend across the Xpert categories?

Figure 14. A regression trend summarizes the Low → Medium → High ordering while accounting for recorded covariates.



Read the figure

Low, Medium, and High are coded 0, 1, and 2. A coefficient is the average three-gene score difference per category step, with an assumption of equally spaced steps. It is not an effect per measured bacterium.

Models progress from cohort adjustment to demographics and then broad composition. Fully adjusted slopes are **0.201 [0.004, 0.398]**, **0.156 [−0.134, 0.446]**, and **0.072 [−0.185, 0.329]**.

Interpret the pattern

The training interval narrowly excludes zero before multiple-testing correction, but its BH-adjusted value is **q = 0.137**. The corresponding test and validation q values are 0.414 and 0.572. None of these three primary trends passes $q < 0.05$.

The present evidence therefore supports a TB-associated expression signal more clearly than a reproducible association with sputum burden. This result does not answer questions about treatment response, progression, radiographic severity, or other clinical outcomes not analyzed here.

Do the selected genes reflect a broader expression program?

Two complementary approaches ask whether coordinated expression extends beyond the selected six genes: relative gene-set enrichment and participant-level expression scores.

Biological theme	Meaning of the gene-set label	Genes in the score after exclusion
Interferon-alpha response	A curated expression-response program associated with type I interferon	91
Interferon-gamma response	A curated expression-response program associated with type II interferon	185
TNF signaling via NF-κB	A transcriptional response associated with this inflammatory signaling axis	157
Inflammatory response	A broader curated inflammatory expression program	132
Neutrophil degranulation	Genes associated with the neutrophil granule-release program	402

1. Gene-set enrichment analysis (GSEA)

The training-only read filter retains 11,279 eligible Ensembl genes, mapped to **11,264 gene symbols**. Within each split, a gene-level regression models VST expression as TB + cohort, then repeats with broad composition adjustment. Genes are ranked by the TB coefficient divided by its HC3 standard error.

GSEA asks whether a predefined set concentrates near the positive or negative end of that list. The normalized enrichment score (NES) standardizes the enrichment against its permutation reference. It is a relative ranking statistic, not a fold change or patient-level difference. [4, 5]

2. Participant-level pathway scores

For each of the five themes, average gene training z scores, excluding the selected six genes, then rescale by that average’s training SD. Regress the resulting participant score on TB + cohort, with and without composition. This directly estimates an aggregate expression difference.

The four focused Hallmark sets contain none of the six selected genes in the pinned definitions. Neutrophil degranulation contains MNDA and LYZ: 404 expressed genes become 402. Removing genes from the entire GSEA background can still alter NES even if a particular set loses no members.

The pathway models do not add age, sex, or HIV. Their covariate scope differs from the fully adjusted clinical models.

Which of the five biological themes remain enriched?

Figure 15. The focused GSEA display compares composition adjustment and removal of the selected genes.



Read the four panels

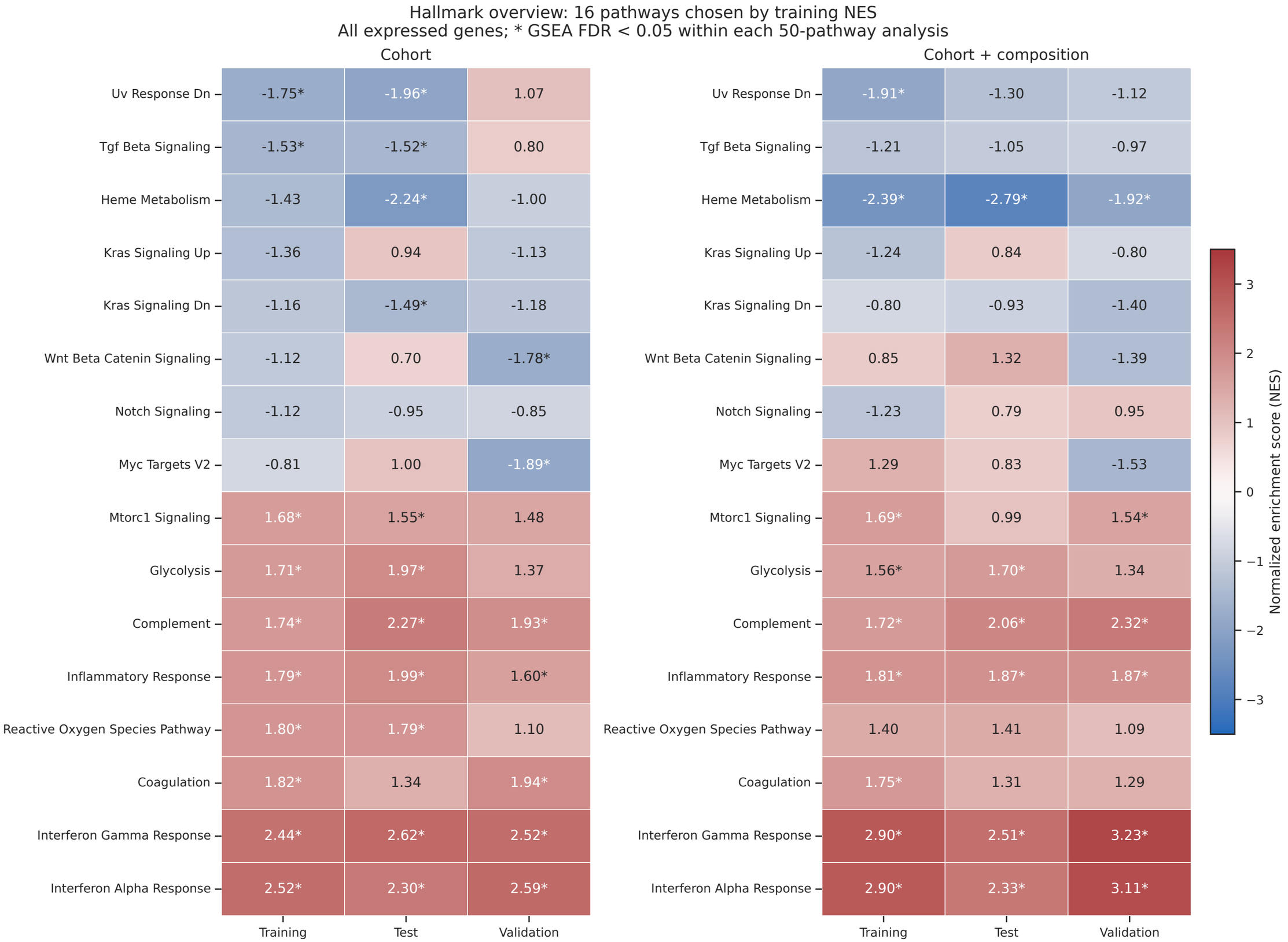
Top versus bottom compares cohort-only with cohort-plus-composition adjustment. Left versus right compares all eligible genes with exclusion of the selected six. Rows are biological themes; columns are splits. Color and numbers show NES; stars indicate GSEA FDR < 0.05.

Interpretation and limits

Positive enrichment persists beyond the selected panel, supporting a broader RNA expression program. These stars use a separate 50-set FDR calculation within each analysis, not one correction over all 600 results. Gene-set permutations do not preserve gene correlations; treat this evidence as exploratory and read it with Figure 17.

What does the broader Hallmark screen show?

Figure 16. The overview displays the eight most positive and eight most negative Hallmark NES values selected using training, cohort-only results.



Read the figure

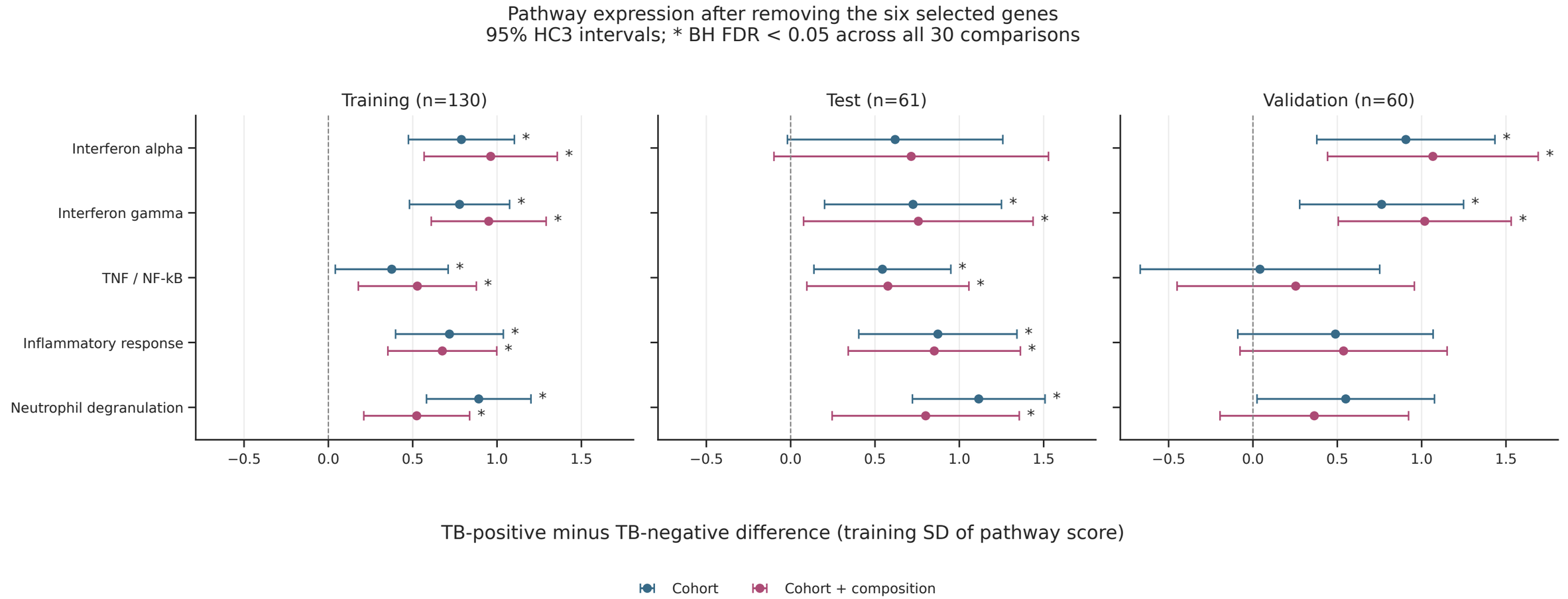
The 16 rows were chosen for display using training only. The **two panels** compare cohort-only and cohort-plus-composition rankings, using **all eligible expressed genes** in both. Positive NES indicates concentration toward higher TB-associated expression; negative NES indicates the opposite. This figure does not show gene exclusion.

Interpret the screen

Interferon and complement show positive enrichment across splits. Other themes change with cohort or adjustment. This is not a validated signature; negative enrichment does not prove pathway suppression. All 50 sets remain in the result table. Stars use the same per-analysis GSEA FDR threshold as Figure 15. The row labels are explained on page 26.

Are the pathway differences also visible at the participant level?

Figure 17. Participant-level scores provide a complementary check on the relative rankings from GSEA.



Read the figure

Dots estimate TB-positive minus TB-negative pathway-score differences in training-SD units; bars are 95% HC3 t intervals. The six selected genes are excluded from all scores. Adjustment sets are shown separately. Stars use BH FDR over **5 pathways × 3 splits × 2 adjustments = 30 tests**.

For interferon-gamma after composition adjustment, the estimates are **0.951 [0.610, 1.291]**, **0.757 [0.077, 1.437]**, and **1.018 [0.506, 1.531]**. All three meet BH q < 0.05.

Interpret the agreement and disagreement

Interferon-gamma has the most consistent support. Interferon-alpha is also supported in validation, but its test interval crosses zero. Validation TNF, inflammatory, and neutrophil score intervals cross zero despite positive GSEA results.

This is not a contradiction: GSEA measures position within a genome-wide rank list; the score measures an aggregate expression difference. Interferon-alpha and gamma sets overlap, so they are not independent pathway discoveries. Neither approach identifies the cell producing the RNA or measures cytokine activity directly.

Optional robustness check: exclude the signature genes

Primary results use the existing manuscript plasma estimates. This additional check asks whether including the six signature genes influences the estimated RNA-source association.

A matched comparison for each participant

Deconvolution estimates which source profiles, and what relative weights, can explain the observed RNA mixture. It uses reference expression patterns from labeled cells to interpret a mixed sample.

Baseline: fit BayesPrism using the available gene counts and the recovered reference.
Excluded: fit the same participant again after removing GBP5, AHNAK, JMJD1C, MNDA, LYZ, and SPI1 from the mixture and reference.

The two fits retain the same reference profiles, random seed, settings, and baseline outlier-gene filter. All six genes survive that baseline filter in these samples, so the exclusion is a real change to the inputs.

The recovered reference has 9,834 cells collapsed into 41 source profiles. Stable Ensembl IDs are harmonized across reference and mixtures, avoiding accidental omission from version-suffix mismatches. The cleaned reference has 29,546 genes; 29,435 overlap the mixtures.

Technical details

BayesPrism combines reference information with the observed counts to estimate a **posterior** distribution: uncertainty after conditioning on the data and model. Version 2.2.3 uses 1,000 initial and 1,000 final Gibbs iterations here. Gibbs sampling repeatedly samples conditional distributions; 500 burn-in iterations are discarded, and thinning of two retains every second subsequent draw. The reference is also updated.

The paired comparison isolates six-gene removal under this reconstructed pipeline. It is not an exact historical reproduction: the original filtering branch and software history remain incompletely recovered. [6]

Which outputs will answer the question?

Fraction stability: compare baseline and excluded source estimates for the same participants. Large changes would indicate direct sensitivity to signature-gene inclusion.

Signal–source correlations: repeat the raw and cohort/TB-adjusted rank associations separately in each split.

Adjusted TB associations: repeat the primary broad-composition regression using excluded-gene fractions, so its coefficients can be compared with the matched baseline.

How to interpret either outcome

If the results barely change, that reduces concern about direct dependence on these six genes. It does not establish independence: correlated genes, reference mismatch, and RNA-release differences can still matter.

If the results change substantially, the current mixture interpretation is sensitive to marker inclusion. Removing genes also changes the RNA denominator slightly. The difference is a sensitivity of estimated RNA contributions, not a change in measured cell numbers.

Status at this update: the previously approved background job continues with eight workers and a 32 GiB memory cap. Completed pairs are checkpointed, but final comparisons are not yet available. This optional run is not required to use, interpret, or report the primary results.

A biological interpretation that matches the evidence

The completed analyses support a TB-associated interferon program, alongside substantial cohort and RNA-source structure. They do not yet separate cell abundance, RNA release, and cell-intrinsic expression.

What is supported

A broader expression program: the interferon-gamma score remains associated with TB beyond the six selected genes under the broad adjustment model in all three splits. This is stronger support for a program-level description than interpreting one selected marker alone.

A mixture-related component: SPI1–MNDA–LYZ tracks myeloid RNA fractions, especially monocyte and neutrophil estimates. Separating those contributions attenuates the training and validation TB coefficients.

Strong cohort structure: broad PCA, UMAP, and K-means agree that cohort and estimated mixture differences are major sources of variation. Reproducible clusters can therefore reflect reproducible population structure.

How this relates to existing TB biology

Published whole-blood work has described interferon-associated and neutrophil-related transcriptional patterns in TB, with both composition and altered gene expression contributing. That provides biological context for these results, but plasma cfRNA is a different material and does not inherit the same cellular interpretation automatically. [7]

The next analyses that could change the interpretation

1. Test the interferon score with finer myeloid adjustment. This can use the existing plasma fractions immediately. The current pathway models still pool myeloid sources; separate monocyte/neutrophil ratios would test the sensitivity of the strongest pathway result.

2. Compare matched plasma with whole-blood estimates. The 60 validation participants already have both assays. They can show whether a plasma pattern parallels whole-blood RNA composition, although both remain inferred RNA contributions.

3. Seek independent cell or tissue evidence. Differential blood counts, flow cytometry, or matched cellular expression would better separate abundance from within-cell responses. Fraction estimates alone cannot resolve this.

4. Freeze a short hypothesis set before a new cohort. Prespecify the interferon score, mixture groups, covariates, and endpoints. Do not label the current clusters as TB subtypes without cohort-independent biological and clinical replication.

5. Add the optional exclusion results when available. Compare the reconstructed baseline with its matched six-gene-excluded fit, inspect diagnostics, and report sensitivity alongside the primary existing-estimate analysis. Primary interpretation does not wait for this run.

Technical details that keep the interpretations aligned

The same participants appear across notebooks, but transformations, outcomes, and adjustment sets are not identical.

Normalization and feature selection

Original model/SHAP and clinical signal: reuse the split-specific DESeq2 VST underlying the completed classifier analysis. The SHAP notebook refits the selected XGBoost pipeline once and verifies its 251 predictions against the existing export.

Patterns and pathways: use a VST reference fitted from training. Eligible genes have at least ten reads in at least 13 training participants; this filter does not use TB labels. Patterns retain the 1,000 highest training-variance genes; pathways use the broader eligible set.

XGBoost: a gradient-boosted ensemble of decision trees. Successive trees improve the fitted prediction rule; nonlinear thresholds allow contributions to depend on other genes. **GFS:** greedy feature selection searches gene subsets. This panel was selected using fivefold cross-validation (CV) within training, scored by AUC. CV repeatedly fits on four folds and evaluates on the fifth. No selection is rerun for this report.

AUC: the area under the receiver-operating-characteristic curve measures how well scores rank a TB-positive participant above a TB-negative participant. It is not the proportion of correct predictions at one cutoff. No figure here uses SHAP magnitude as an AUC substitute.

Why adjustments are conditional

The primary composition model has four groups: seven myeloid labels, five lymphoid labels, erythroid/platelet, and the other 27 labels. It uses three log ratios to “other.” The clinical myeloid sensitivity uses six groups and five ratios, preserving that denominator.

Clustering and statistical uncertainty

K-means: assigns participants to centers to reduce within-group squared distances. **Silhouette:** compares within-group closeness with the nearest alternative group. **ARI:** chance-adjusted agreement between two group assignments; 1 is identical and 0 is approximately chance agreement.

Bootstrap: resample participants with replacement, refit the clustering, and compare labels. The 100 bootstrap repetitions here keep earlier transformations fixed. They do not measure uncertainty from the entire analysis pipeline.

Residual degrees of freedom: participant count minus the number of fitted regression coefficients. Small strata and many covariates can leave limited information for estimating uncertainty.

BH correction: the Benjamini–Hochberg procedure adjusts p values within a declared family. The clinical primary TB family has three tests; the burden family has three separate tests; the pathway-score family has 30. Sensitivity analyses do not become confirmatory because an individual interval excludes zero.

GSEA scope: 50 Hallmark sets plus one Reactome set were offered; the pancreatic beta-cell Hallmark set had only 12 eligible genes and failed the 15-gene minimum. Thus 50 sets were analyzed per run. Across three splits, two adjustments, and two exclusion settings, 600 result rows were saved.

Fraction uncertainty is not propagated through the downstream regressions. Posterior variability from BayesPrism is not, by itself, proof that the sampler or reference-update optimizer converged.

What do the pathway names in Figure 16 mean?

These are curated gene-set labels from MSigDB. They name expression programs used to interpret a rank list; they do not establish that a specific signaling event occurred in these participants. [4]

The first eight rows

UV response Dn: genes reduced after ultraviolet exposure in the source experiments. “Dn” belongs to the reference label; it does not mean the current participants had UV exposure or that the current NES must be negative.

TGF beta signaling: an expression response associated with transforming growth factor beta, a signaling system involved in cell state and tissue remodeling.

Heme metabolism: genes involved in heme synthesis and erythroid differentiation. Heme is the iron-containing component of hemoglobin and other proteins; mixture differences can influence this program.

KRAS signaling Up / Dn: sets induced or reduced by KRAS activation in source experiments. KRAS is a signaling protein. “Up” and “Dn” identify two reference sets, not the sign of the TB comparison and not evidence of a KRAS mutation.

Wnt beta-catenin signaling: a developmental and cell-state signaling program. Beta-catenin helps regulate transcription in this pathway.

Notch signaling: a cell-communication program involved in differentiation and cell fate.

MYC targets V2: a second curated subset of genes regulated by the MYC transcription factor, which influences cellular growth. “Targets” means expression-regulatory targets, not drug targets.

The last eight rows

mTORC1 signaling: an expression response associated with the mTOR complex 1 nutrient/growth signaling system.

Glycolysis: a program associated with breaking down glucose to obtain energy and metabolic intermediates.

Complement: an immune-defense protein system involved in recognition, clearance, and inflammatory responses.

Inflammatory response: a broader collection of genes associated with inflammation, rather than one specific cytokine or cell type.

Reactive oxygen species pathway: an expression response to chemically reactive oxygen-containing molecules and oxidative stress.

Coagulation: genes associated with clotting and its related responses. An RNA enrichment is not a direct measurement of a clot or clotting activity.

Interferon-gamma response: a curated transcriptional response associated with a type II interferon immune signal.

Interferon-alpha response: a curated response associated with type I interferon. Its members overlap the gamma set; neither set uniquely measures the corresponding cytokine concentration.

A transcription factor is a protein that regulates gene expression. A signaling pathway is a chain of cellular communication events. Here those biological concepts are represented by gene lists, not direct biochemical measurements.

Sources and reproducibility

All plotted figures are the existing notebook exports, embedded as vectors so that labels remain sharp when zoomed. No model or biological analysis was rerun to create this PDF.

Local notebooks and evidence

Scripts_For_Manuscript/
gene_interactions_interpretation.ipynb
gene_interactions_patterns.ipynb
gene_interactions_cohort_clinical.ipynb
gene_interactions_pathways.ipynb
gene_interactions_deconvolution.ipynb

Figure_Output/Gene_Interactions/focused/
Each figure page identifies its exact PDF stem. Corresponding PNGs and numerical CSVs are stored alongside them. Key evidence includes myeloid_signal_correlations.csv, myeloid_tb_associations.csv, patterns_cluster_stability.csv, cohort_clinical_tb_adjusted_associations.csv, cohort_clinical_tb_separate_myeloid_adjustment.csv, pathway_enrichment.csv, and pathway_score_tb_associations.csv.

Files_For_Manuscript/
Raw counts, participant metadata, discovery_cell_types.tsv, and cto_wb.txt are the original local inputs. Training-reference expression, eligibility lists, pinned pathway definitions, and deconvolution preparation caches are in the existing gene_interactions_reference directory.

Notebook-specific provenance JSON files record source hashes, analysis settings, package versions, and exact exported artifacts. Figures and tables were checked for consistency during report assembly. Primary composition results use the existing manuscript plasma estimates; pending optional gene-exclusion outputs are not incorporated.

Prepared 21 October 2026. The report is a research interpretation, not a clinical performance or diagnostic recommendation.

Primary methods and biological context

[1] Original cfRNA TB study. Nature Communications (2024), DOI: 10.1038/s41467-024-49245-6. Study context, assay distinction, and Xpert Ultra labeling.
Read the study.

[2] XGBoost prediction API and TreeSHAP documentation. Native contribution/interaction outputs and tree-path-dependent attribution.
XGBoost API · TreeSHAP documentation.

[3] UMAP documentation. Interpretation of clustering and limits of density/gap preservation.
Using UMAP for clustering.

[4] MSigDB. Pinned human release 2026.1; Hallmark and Reactome definitions.
Release files · Hallmark sets and definitions.

[5] Subramanian et al. (2005). Gene set enrichment analysis, PNAS, DOI: 10.1073/pnas.0506580102.
Primary methods paper.

[6] Chu et al. (2022). BayesPrism, Nature Cancer, DOI: 10.1038/s43018-022-00356-3.
Official implementation and usage guidance.

[7] Berry et al. (2010). Interferon-associated whole-blood transcription in TB, Nature, DOI: 10.1038/nature09247.
Primary study.

How each analysis was made

Read these pages alongside the original figures and interpretations on pages 1–27, which are preserved in full. The explanation moves from the biological measurement to the calculation, its assumptions, and the meaning of the plotted quantity.

Original report	Question answered by the additional methods	Read here
Figures 1–3 · pp. 5–7	How counts become model inputs; how the six-gene XGBoost model was selected and fitted.	pp. 29–30
Figures 1–2 · pp. 5–6	Where SHAP comes from; its reference value; a real participant’s expression → SHAP → probability calculation.	pp. 31–32
Figure 3 · p. 7	How pairwise SHAP is calculated, why the factor of two is needed, and what the heatmap averages.	pp. 33–34
Figures 4–6 · pp. 8–10	How RNA-source fractions and the SPI1–MNDA–LYZ score were assembled; correlations and adjusted TB differences.	pp. 35–36
Figures 7–10 · pp. 11–14	Which genes entered the heatmap, PCA, UMAP, and clustering; what was fitted only in training.	pp. 37–38
Figures 11–14 · pp. 15–18	How cohort, composition, technical depth, and Xpert categories enter the clinical models.	pp. 39–40
Figures 15–16 · pp. 20–21	How genes were ranked, how enrichment was tested, and what NES and the reported FDR mean.	pp. 41–42
Figure 17 · p. 22	How participant-level pathway scores were calculated and why they answer a different question from GSEA.	p. 43
Optional check · p. 23	How the paired gene-exclusion deconvolution is designed and what it can establish after completion.	p. 44
All analyses	What was verified, what remains conditional, and where the methods and implementation are documented.	pp. 45–46

A consistent way to read the methods

A **measurement** is an observed count or recorded clinical field. A **transformation** changes its scale. A **model** combines inputs according to fitted rules. A **summary statistic** condenses the resulting values. These are distinct steps, with different interpretations.

What was done for this edition

The additions trace the executed notebooks, settings, and saved numerical outputs. No feature-selection search, model fitting, or deconvolution was repeated to write this companion. The worked SHAP example is reconstructed from existing values.

From read counts to the expression axes

Expression values are calculated before SHAP. SHAP then explains what the fitted model does with those values; it does not estimate expression.

1 · Start with the three count matrices

Each entry K_{ig} is the RNA-sequencing read count for participant i and gene g in counts.train.tsv, counts.test.tsv, or counts.val.tsv. Each file contains 56,634 genes. Sample IDs align counts to metadata; the disjoint splits contain 130, 61, and 60 participants.

2 · Correct for sample-wide count scale

The GFS/SHAP analysis reproduces the paper-style normalization separately within each split. For genes positive in every participant of that split, calculate a geometric reference $r_g = \exp[\text{mean}_i(\ln K_{ig})]$. The participant's size factor is:

$$\mathbf{s_i = \exp\{\text{median}_g[\ln K_{ig} - \ln r_g]\}}$$

Then $q_{ig} = K_{ig} / s_i$.

The size factor describes the relative count scale across many genes. It is not the measured amount of circulating RNA, and it does not correct for gene length.

3 · Stabilize count-dependent variance [M4]

Counts have larger variability at larger means. The DESeq2 parametric variance-stabilizing transformation (VST) compresses that relationship using the fitted dispersion trend:

$$\mathbf{x = \log_2\{[1 + b + 2Aq + 2\sqrt{(Aq(1 + b + Aq))}] / (4A)\}}.$$

Here q is the size-factor-normalized count, A is asymptDisp, and b is extraPois. The saved (A, b) values are training (0.18210449, 3.29320823), test (0.16988295, 3.09435352), and validation (0.15396245, 3.38438462).

What the VST scale means

The dispersion trends were fitted with an intercept-only design, without TB labels, within each split. This is a blind transformation, reconstructed with DESeq2 1.30.1; the article reported 1.34.0. It uses varianceStabilizingTransformation, not the faster vst approximation.

VST values are neither raw counts nor TPM, and they are not simply $\log_2(\text{count} + 1)$. A one-unit difference is a transformed-expression difference; it is not generally a twofold change in RNA abundance.

4 · Standardize for the fitted classifier

The final pipeline fits a mean imputer and StandardScaler on training. For each gene, $z_{ig} = (x_{ig} - \mu_{g,\text{train}}) / \sigma_{g,\text{train}}$, where σ uses the population-SD convention (divide variance by n). The same training μ and σ transform test and validation. There are no missing values in the current exported input, so imputation changes nothing.

SHAP summary colors use z. A color at +1 means one training SD above that gene's training mean. **SHAP dependence x-axes use x**, the VST values before this standardization. The six-gene model receives z in both cases.

Two normalization paths in this report

The SHAP and original myeloid-score analyses read gfs/expression_vst.csv, with independent VST references in each split. The expression-pattern and pathway notebooks instead read expression_training_vst.npz, freezing the training reference and dispersion trend for the holdouts. These paths should not be treated as identical inputs.

Split-specific VST uses the expression distribution of the whole evaluation batch. It reproduces the comparison's preprocessing, but it is not a fully frozen procedure for predicting one new sample in isolation. This also matters when comparing expression and SHAP patterns across splits.

Sources: gene_interactions.ipynb · variance_stabilize; interpretation and follow-up notebook input cells

Which model is being explained?

All three SHAP figures explain one final six-gene XGBoost model. The genes were selected together by a training cross-validation search.

1 · Define the candidate genes

Training-only differential expression used DESeq2 with design ~ TB status, comparing TB-positive with TB-negative participants. Genes with adjusted $P < 0.05$ and mean normalized count > 100 formed the candidate pool: 142 genes. The article's reported pool of 113 remains unresolved; this is the audited 142-gene extension.

2 · Compare subsets within training

Five stratified folds, shuffled with seed 87, preserve the approximate TB-positive/negative balance. Each fold holds out 26 of the 130 training participants. For a candidate subset, fit imputation, scaling, and XGBoost on four folds, predict the fifth, and average the five ROC-AUCs.

ROC-AUC measures how often a randomly chosen TB-positive participant receives a higher score than a negative participant, with half credit for ties. It assesses ranking, not probability calibration.

Forward search begins at AUC 0.5, tries each unused gene, and accepts the best addition if its mean-AUC improvement exceeds 10^{-12} (a numerical threshold). Ties use the lowest absolute mean Pearson correlation with accepted genes, then candidate order. The later refinement also tries removal and replacement.

3 · Retain the resulting panel

GBP5, AHNAK, JMJD1C, MNDA, LYZ, SPI1 were retained, in that order. Six was the search result, not a forced size. Refinement returned the same panel. Mean selection-CV AUC was 0.93738; adding SPI1 increased it by about 0.00483. A later STAT1 addition did not improve it.

4 · Fit the final model once

The interpretation notebook reads the completed selection results and refits that pipeline on all 130 training participants. XGBoost 3.0.5 uses **100 trees, maximum depth 2, learning rate 0.05, histogram split search, binary logistic loss, and seed 87**. There is one numerical thread.

Other settings retain library defaults, including L2 leaf-weight penalty 1, L1 penalty 0, minimum child Hessian weight 1, and full row/column sampling. There are no sample weights, class reweighting, or biological interaction constraints. All 100 rounds are fitted; no holdout evaluation set or early stopping is used. [M3]

The six standardized gene values are the only model inputs. Cohort, RNA-source fractions, read depth, age, and clinical burden are not classifier covariates. A label is needed for fitting and evaluation, but not for calculating a fitted-model explanation.

5 · Explain the same model in all splits

Final-model AUC is 0.99123 in training, 0.91285 in test, and 0.97200 in validation. Training explanations are in-sample, not out-of-fold. Prediction checks match the previously exported forward and refined models.

Limits of the selection score

The DE screen uses all training outcomes before inner CV, and the training VST is also prepared before the folds. Those steps are not nested within CV. Therefore 0.93738 is a criterion used to choose features, not an unbiased estimate of the entire discovery procedure.

Holdout labels do not enter the programmed search, but these holdouts have been inspected repeatedly during exploratory work. SHAP describes the fitted predictor; it does not correct selection bias, establish calibration, or guarantee prospective performance.

Sources: gene_interactions.ipynb · CV GFS and revisions; focused/gfs/settings.json and saved selection results

How SHAP assigns a contribution to a gene

SHAP—SHapley Additive exPlanations—uses a game-theoretic idea: share a prediction among its input features according to their average incremental contribution. [M1–M3]

Begin with the scale being explained

For this binary classifier, the raw model output f(z) is the log odds of TB-positive status. Probability is p = 1 / [1 + exp(−f)]. SHAP decomposes that raw output:

$$\mathbf{f(z_i) = \varphi_0 + \Sigma_j \varphi_{ij}}.$$

φ_0 is the reference value, and φ_{ij} is gene j's contribution for participant i. A positive contribution increases predicted TB log odds relative to the reference; a negative contribution decreases them. A contribution of +1 is not a one-unit increase in probability or a regression slope.

What does "incremental" mean?

Let F be the six input genes and S a subset already revealed. The value $v_i(S)$ is the tree-based expected raw prediction when the participant's values for S are known and the remaining genes are integrated out. Revealing gene j changes this value by:

$$\mathbf{v_i(S \cup \{j\}) - v_i(S)}.$$

SHAP averages this change over all possible feature orderings, so a gene receives credit both when it enters early and when it enters after other genes:

$$\mathbf{\varphi_{ij} = \Sigma_{S \subseteq F\setminus\{j\}} w(S) \times [v_i(S \cup \{j\}) - v_i(S)]},$$

where $w(S) = |S|!\,(6 - |S| - 1)! / 6!$.

Here |S| is the number of revealed genes and ! denotes a factorial. A missing feature in this calculation is **unknown to the explanation**; it is not set to zero and is not a simulated gene knockout.

How our implementation evaluates this

The notebook calls XGBoost's native booster with pred_contribs=True and approx_contribs=False. TreeSHAP computes these weighted contributions efficiently from the fitted trees. It does not refit a model for every subset, and it does not use sampled KernelSHAP explanations or a separate background dataset.

At a split on a known gene, the participant follows its observed branch. At a split on an unknown gene, the calculation averages over branches using stored node cover. In XGBoost 3.0.5 this cover is the **sum of training-loss Hessians**, rather than simply the number of training participants. [M3]

A Hessian here measures the curvature of the loss used to fit a tree. For unweighted logistic loss its observation-level weight is approximately p(1−p) at that boosting stage. Thus the tree reference can weight training participants unequally even without explicit sample weights.

What is the baseline in these figures?

It combines the initial raw intercept (estimated from training labels) and the cover-weighted expected outputs of the fitted trees. It is constant across all 251 explanations: $\mathbf{\varphi_0 = 0.25357887}$ **log odds**, corresponding to probability 0.56305719.

This is not the prediction at mean gene expression and not the arithmetic mean training prediction. For comparison, the saved mean training logit is about 0.42249. Applying the logistic function to a mean logit also need not equal the mean probability.

What "exact" establishes

The attribution is exact for this fitted model and tree-path reference rule, up to numerical precision. It is not an exact biological effect or a guaranteed population conditional expectation. Correlated inputs affect how credit is shared. Histogram tree fitting itself uses approximate split search.

A real participant: expression → SHAP → probability

This example is the participant with the median predicted probability among the 61 test samples, selected without using the observed TB label. Sample ID: R2D2_V242_combined_61bp. All numbers below come from the saved explanation table.

Gene	VST expression x	Training mean μ	Training SD σ	Model input z	SHAP φ (log odds)
GBP5	8.75888	7.53827	1.05277	+1.15943	+1.60763
AHNAK	9.51950	9.83166	0.49853	-0.62616	-0.04532
JMJD1C	6.62853	6.87074	0.35110	-0.68985	+0.60916
MNDA	7.41500	8.16171	0.72409	-1.03123	-0.86911
LYZ	9.05637	8.77479	0.86315	+0.32621	-0.03055
SPI1	7.32172	7.84263	0.60920	-0.85509	-0.00279

1 · Reproduce one expression input

For GBP5, z = (8.758882 − 7.538272) / 1.052768 = **1.159430**. Thus this participant's GBP5 value is about 1.16 training SDs above the training mean. Its VST value came from the test split's count normalization; its centering and scale came from training.

2 · Read the fitted-model contribution

TreeSHAP assigns GBP5 a contribution of **+1.607632 log odds**. This is calculated by tracing the fitted trees and averaging incremental contributions over coalitions. It is not z multiplied by a coefficient; XGBoost is a nonlinear tree ensemble.

JMJD1C has below-average expression (z = −0.68985) but a positive SHAP contribution (+0.60916). An expression value and a contribution are different quantities. Their relationship depends on the fitted tree rules and the other gene values.

3 · Add contributions on the log-odds scale

The six unrounded contributions sum to **1.26901078**. Add the baseline: **0.25357887 + 1.26901078 = 1.52258965**.

4 · Convert once to probability

p = 1 / [1 + exp(−1.52258965)]
= 0.82091950 ≈ 82.09%.

The saved model probability is 0.82091945; the tiny difference is numerical precision. Do not apply the logistic function separately to each gene's contribution and add the resulting values: probabilities do not add this way.

The participant was recorded as TB-positive. That label is useful for checking the prediction, but it did not enter this SHAP calculation. This is a worked calculation, not evidence that the probability is perfectly calibrated or that any gene caused TB.

How SHAP of gene pairs was calculated

A pairwise SHAP value asks whether the fitted model's attribution for two genes jointly differs from their separate contributions. It uses the same six-gene model, without training separate pair models.

The joint increment

For genes j and k and a revealed subset S containing neither gene, first calculate a second difference:

$$\Delta_{jk}(S) = v(S \cup \{j,k\}) - v(S \cup \{j\}) - v(S \cup \{k\}) + v(S).$$

If the two genes contribute additively under this explanation rule, this difference is zero. Nonzero values describe model nonadditivity after averaging over the other possible revealed genes.

For the six-feature model, each off-diagonal entry is the weighted sum:

$$\Phi_{i,jk} = \Sigma_{S \subseteq F\setminus\{j,k\}} w_2(S)\, \Delta_{jk}(S),$$

where $w_2(S) = |S|!\,(6 - |S| - 2)! / [2(6 - 1)!]$. [M2]

The output is a matrix for each participant

XGBoost's pred_interactions=True, approx_contribs=False returns a 7 × 7 matrix: six genes plus the bias. After checking row sums, the notebook removes the bias row and column and saves the signed 6 × 6 matrix.

The diagonal $\Phi_{i,jj}$ contains the SHAP main effect. Off-diagonal cells contain pair allocations, with $\Phi_{i,jk} = \Phi_{i,kj}$ up to floating-point precision. Each ordinary gene SHAP value equals its row sum:

$$\varphi_{ij} = \Phi_{i,jj} + \Sigma_{k \neq j} \Phi_{i,jk}.$$

Why the heatmap multiplies by two

One unordered pair occupies two symmetric matrix cells. Each cell carries half the pair's complete contribution. Therefore:

$$f(z_i) - \varphi_0 = \Sigma_j \Phi_{i,jj} + 2\,\Sigma_{j<k} \Phi_{i,jk}.$$

For each split s, the figure averages the magnitude of that complete pair contribution across participants:

$$P_{jk,s} = (1/n_s)\,\Sigma_{i \text{ in } s} |2\Phi_{i,jk}|.$$

The absolute value is taken **before** averaging. This prevents positive and negative participant-level contributions from cancelling. It also removes direction: a bright cell does not say whether the pair raises or lowers predicted TB probability.

Continue the worked example

For the participant on page 32, the AHNAK–MNDA off-diagonal entry is approximately −0.088710. The full unordered-pair contribution is about **−0.177420 log odds**. For GBP5–MNDA it is about **+0.132684**. Different pairs can contribute in different directions for the same person.

Do not add pair effects to ordinary SHAP again. Ordinary gene SHAP already includes its assigned pair contributions. Use either the six ordinary contributions, or the diagonal main effects plus the doubled unique pairs.

What this does not test

The heatmap has no null model, P value, confidence interval, or FDR correction. It is a description of the fitted model. It does not demonstrate protein binding, regulatory interaction, or biological epistasis. Correlated expression and changing RNA sources can also contribute to these patterns.

Sources: gene_interactions_interpretation.ipynb · pred_interactions and pair_strength; signed interaction arrays [M2–M3]

How the three SHAP figures were assembled

The model and explanation algorithm are fixed. Each figure displays a different summary of the same participant-level contributions.

Figure 1 · Summary dots, p. 5

One dot is one participant's contribution for one gene. Its horizontal coordinate is φ_{ij} in log-odds units. Vertical offsets are uniform random display jitter (seed 87), used to separate overlapping points; they are not another biological variable or a density estimate.

Dot color is that gene's training-standardized expression z. Colors saturate outside −2 to +2, but the expression and SHAP values are not truncated. Training, test, and validation share the same horizontal SHAP scale.

Genes are ordered by their **training mean absolute SHAP**: $I_{j,s} = (1/n_s) \Sigma_{i \text{ in } s} |\varphi_{ij}|$. The other panels retain that order to support direct comparison. This average includes every participant in the split without class balancing.

A larger mean absolute value means a larger average contribution to this model's predictions. It is not a significance score, independent diagnostic accuracy, or proof of a larger causal biological role.

Figure 2 · Expression dependence, p. 6

For each gene, the x-axis is the original VST expression and the y-axis is that same participant's SHAP contribution. Gray circles, blue squares, and gold triangles distinguish training, test, and validation. Each gene has its own axis limits.

No regression curve is fitted, and no gene expression is experimentally or computationally altered. Steps and plateaus can arise from tree thresholds. Vertical variation at similar expression can reflect the other selected inputs and the attribution rule.

Figure 3 · Pairwise magnitude, p. 7

There are 15 unique pairs among six genes. The heatmap shows only the lower triangle, excluding the diagonal main effects and duplicated pair entries. It uses the training SHAP gene order and one common color scale across splits.

Each cell is mean $|2\Phi_{jk}|$ in log-odds units, as derived on page 33. The signed participant-level arrays are retained in xgboost_cv_shap_interaction_values.npz, so direction can be examined separately.

What cross-split differences mean

The three panels apply one fitted model to different participants. A larger SHAP or pair magnitude in validation may reflect different expression distributions, class proportions, RNA sources, cohorts, or split-specific preprocessing. It is not evidence that the model retrained itself or acquired a new mechanism in validation.

GBP5–AHNAK has the largest mean absolute pair contribution in each split: approximately 0.1368, 0.1522, and 0.2663 log odds in training, test, and validation. These are descriptive magnitudes; the heatmap does not test whether the differences are statistically reliable.

Why SPI1 can be selected but look unimportant

Feature selection compares predictions from five separately fitted fold models. SHAP explains the final model refitted on all training participants. A gene can improve the average fold score yet receive little attribution after the full-data refit; correlated genes may carry similar information.

Numerical consistency checks

For all 251 saved explanations, SHAP plus baseline reconstructs probability within 1.86×10^{-7}. Interaction row sums reproduce ordinary SHAP within 3.52×10^{-7}. These checks validate the accounting, not causal interpretation or predictive generalization.

Sources: xgboost_cv_shap_values.csv, importance.csv, interactions.csv, interaction_values.npz, checks.csv

From plasma RNA sources to the myeloid signal

Figures 4–5 compare two summaries derived from the plasma RNA assay: a fixed expression score and estimated contributions from reference cell types.

Reuse the existing manuscript estimates

discovery_cell_types.tsv contains 268 participants and 41 RNA-source columns. All 251 participants in the model splits have matching sample IDs. The analysis checks unique IDs, finite nonnegative fractions, and a per-participant sum of one before joining them to expression and metadata.

These are **relative contributions to the modeled plasma RNA mixture**. They are not measured numbers of circulating cells. Cell abundance, cell death, release of RNA, and reference-profile differences can all influence the relationship between RNA fractions and actual cell numbers.

cto_wb.txt is a distinct whole-blood RNA-source table for 60 participants. It is not substituted for plasma estimates in these figures. The optional gene-exclusion rerun is likewise not an input to the current primary results.

Define the grouped fractions once

Myeloid sums monocyte, macrophage, neutrophil, conventional dendritic cell, myeloid progenitor, basophil, and mast-cell reference contributions.

Lymphoid sums NK cell, B cell, T cell, plasmablast, and thymocyte. **Erythroid/platelet** sums erythrocyte/erythroid progenitor and platelet. **Other** contains the remaining 27 labels. These nonoverlapping groups exhaust all 41 sources.

Construct the three-gene score

$$S_i = (z_{SPI1,i} + z_{MNDA,i} + z_{LYZ,i}) / 3.$$

The z values are the training-standardized GFS/SHAP inputs defined on page 29. All three weights are fixed at +1/3; none is fitted in this analysis. The composite is not rescaled to SD = 1. A difference of 0.3 is 0.3 mean gene-z units, not 30% expression or 0.3 composite SDs.

Figure 4 · Raw and adjusted correlations

Each participant’s x coordinate is 100 × myeloid fraction and its y coordinate is S. Colors identify TB status and shapes identify cohort. All three splits are analyzed separately.

Spearman ρ is Pearson correlation applied to ranks, with tied values receiving average ranks. It describes whether people high in one quantity also tend to be high in the other; it does not estimate a change in expression per cell.

For the **partial rank correlation**, rank S and the myeloid fraction within the split. Regress each ranked vector on an intercept, TB status, and cohort indicators. Correlate the two residual vectors using Pearson correlation. The resulting number describes association remaining after those linear effects on ranks are removed.

Validation contains only TBSQUARE, so no varying cohort term can be estimated there. The remaining adjustment is for TB status. This does not provide between-cohort validation of a common slope.

Figure 5 · Source-specific correlations

For each split, calculate raw Spearman correlations between each of four expression summaries (S, SPI1, MNDA, LYZ) and each of eight fraction summaries (myeloid total and seven sources). Display those coefficients on a shared −1 to +1 color scale.

These heatmap cells are **unadjusted correlations**. They have no significance threshold or multiplicity correction. The rows are related: seven components sum to the total, and all fractions belong to one constrained mixture.

What the association can establish

Concordance supports the interpretation that the expression score tracks an estimated myeloid RNA contribution. It does not identify a unique source or independently validate deconvolution, because both axes are derived from the same RNA assay and can share marker information.

Sources: interpretation notebook · signal_expression, rna_groups, correlation_rows; discovery_cell_types.tsv

How the adjusted associations and intervals work

Figure 6 models a continuous expression outcome. The coefficient of TB is a conditional expression difference, not a disease probability or classifier-performance metric.

Represent composition without a redundant total

All fractions sum to one. Including every raw fraction with an intercept would create a linear dependency. The primary models instead use three additive log ratios (ALRs):

$$L_M = \ln(\text{myeloid} / \text{other})$$
$$L_L = \ln(\text{lymphoid} / \text{other})$$
$$L_{EP} = \ln(\text{erythroid-platelet} / \text{other}).$$

These describe the four-part composition using three coordinates. All grouped fractions are positive in the current data, so no pseudocount or zero replacement is applied. One log-ratio unit is an e-fold change in the source/reference ratio; doubling is ln(2) units.

Fit a separate model within each split

For outcome Y, the primary initial model is:

$$Y_i = \beta_0 + \beta_{TB} TB_i + \beta_C C_i + \gamma_M L_{M,i} + \gamma_L L_{L,i} + \gamma_{EP} L_{EP,i} + \varepsilon_i.$$

TB is 1 for positive and 0 for negative. C is a cohort indicator in training/test, which contain GHL and R2D2. Validation has one cohort and no such contrast. Y is either the three-gene score or one gene's training z score.

Ordinary least squares (OLS) chooses coefficients that minimize the sum of squared residuals ε. β_{TB} estimates the positive-minus-negative expression difference at the same modeled covariate values. The association regressions are fitted anew in each split; they are not classifiers trained in one split and tested in another.

What the four adjustment rows compare

Figure 6 shows TB alone; TB + cohort; TB + cohort + the three broad ALRs; and TB + cohort + five finer ALRs. The last model separates erythroid, platelet, and endothelial sources while still pooling myeloid sources. Its Other denominator therefore changes. Age, sex, and HIV are absent from these initial models.

HC3: allow residual variance to differ

HC3 leaves the fitted coefficients unchanged but adjusts their estimated uncertainty for unequal residual variance and leverage. Leverage h_{ii} describes how unusual participant i's covariate values are relative to the design. [M5]

With design matrix X, residual e_i, and participant covariate column vector x_i, the covariance estimate is:

$$V_{HC3} = (X'X)^{-1} \times [\Sigma_i x_i x_i' e_i^2 / (1 - h_{ii})^2] \times (X'X)^{-1}.$$

The standard error is the square root of the corresponding diagonal element. The plotted interval is:

$$\beta \pm t_{0.975,n-p} \times SE_{HC3},$$

where n is participants, p is fitted parameters including the intercept, and t is the Student-t cutoff. The notebooks explicitly request residual-degrees-of-freedom t inference. These are individual 95% intervals, not simultaneous bands.

What uncertainty is included?

The interval describes participant-level residual variation under the regression model and working independence assumption. It is not a cohort-clustered or bootstrap interval. It does not propagate uncertainty from feature selection, normalization, deconvolution, or choice of model.

The initial analysis fits 48 models (3 splits × 4 expression outcomes × 4 adjustment sets), without a multiple-testing correction. The later clinical and pathway analyses have their own explicitly defined correction families.

How adjustment changes the interpretation

A remaining coefficient is a conditional association, not proof of a cell-intrinsic response. Attenuation is not a measured percentage mediated by composition. Reference error, shared RNA information, correlated ratios, and unmeasured variables can change either the estimate or its precision.

Sources: interpretation notebook · make_design and TB associations; statsmodels HC3/t inference [M5]

How the expression heatmap and PCA were made

Figures 7–8 describe expression similarity. They use a training-reference VST cache, which differs from the split-specific VST behind the SHAP model.

Freeze normalization to training

The cache uses training geometric reference means and the training dispersion coefficients from page 29 for all participants. For a held-out participant, only that participant's counts determine its size factor; zero counts are omitted from its reference-ratio median. Neither test nor validation updates the reference.

Input and cache SHA-256 hashes, unique sample IDs, finite values, and metadata alignment are checked before loading. Thus the saved expression source is explicit and reusable.

Figure 7 · Order a six-gene display

For GBP5, AHNAK, JMJD1C, MNDA, LYZ, and SPI1, fit StandardScaler on the 130 training participants and apply it unchanged to all 251. Colors show these training-z values, saturated at ±3 for display.

Columns are ordered using **average-linkage hierarchical clustering** with Euclidean distance across the six genes. Euclidean distance is the square root of the summed squared differences between standardized expression vectors. Average linkage joins groups using the average distance between their members. Gene rows retain their fixed order.

All participants enter this descriptive column ordering; it is not a trained assignment rule. Distances use unclipped values. Annotation strips show TB status, cohort, and split. This is not the KMeans grouping in Figure 10. Because these genes were previously selected for TB prediction, this display cannot independently validate their selection.

Choose the broader PCA genes

A gene is eligible if it has at least 10 reads in at least 13 training participants (10% of 130, rounded up): 11,279 genes pass. Among them, retain the 1,000 with greatest training VST variance. No TB association P value enters this broad filter.

Variance ranking uses sample variance (ddof = 1). StandardScaler then uses the training mean and population SD (ddof = 0). The gene list and scaler are frozen for test and validation; the common variance denominator does not change the ranking.

Figure 8 · Learn linear directions of variation

Principal component analysis (PCA) finds orthogonal weighted combinations of genes that explain the greatest training variance. A six-component PCA is fitted to the six-gene matrix; a separate 20-component PCA is fitted to the 1,000-gene matrix.

Both fits use all 130 training participants, including TB-negative participants, and a full singular-value decomposition. Held-out participants are projected using the same gene list, scaler, PCA center, and loadings. No TB labels enter the PCA fitting.

In matrix form, scores are (Z − training PCA mean) × V, where Z is standardized expression and columns of V are principal-component loadings. The displayed coordinates are unitless linear combinations, not expression of one gene.

Broad PC1 and PC2 explain about 61.8% and 5.4% of standardized training variance; the first 20 explain 86.3%. Six-gene PC1 and PC2 explain about 47.3% and 19.6%. These percentages are not proportions of TB risk explained.

Overlay variables after fitting

Each PCA row repeats the same coordinates four times, colored by TB status, cohort, myeloid RNA fraction, or $\log_{10}$ assigned reads (metadata field ftcount). Marker shapes distinguish the three splits.

There is no cohort, composition, or read-depth correction before PCA. The overlays help identify what the dominant directions track; they do not remove those influences.

Why this matters for later analyses

UMAP and KMeans use all 20 broad PCs, not just the two displayed axes. PCA whitening is disabled, so components retain their different variances. The later Euclidean distances therefore give more influence to larger-variance directions.

Sources: gene_interactions_patterns.ipynb · heatmap, training_features, StandardScaler, PCA; training VST cache

How UMAP and patient groups were evaluated

Figure 9 provides a neighborhood-oriented view. Figure 10 uses a separate clustering algorithm in the original 20-PC representation, rather than drawing groups on the UMAP.

Figure 9 · Fit the UMAP display in training

UMAP constructs a neighborhood graph in the input representation and seeks a low-dimensional arrangement that preserves those relationships. Here the inputs are the 20 broad PC scores. The primary settings are 15 neighbors, min_dist = 0.3, Euclidean distance, and random/transform seed 87. [M6]

The reducer is fitted only on 130 training participants. Test and validation are transformed separately with the fitted reducer and do not move the training points. No TB labels supervise the embedding.

UMAP axes have no intrinsic biological unit. Orientation, apparent island size, and distance between remote islands should not be interpreted like calibrated biological distances. Repeated panels overlay TB, cohort, myeloid fraction, and assigned reads.

Check local-neighborhood sensitivity

The notebook recomputes the embedding with 10 or 30 neighbors, and with seed 88 at 15 neighbors; min_dist remains 0.3. **Trustworthiness** evaluates whether apparent 2-D neighbors were also nearby in the 20-PC space, penalizing false neighbors according to their original distance ranks. The alternative scores are exported; Figure 9 shows the primary embedding.

The score uses 10 neighbors and is evaluated within each split. Primary values are about 0.943, 0.941, and 0.857 for training, test, and validation. It assesses local representation quality, not classifier accuracy, preservation of all global distances, or proof of subtypes.

Figure 10 · Fit KMeans to training TB cases

Keep the 73 TB-positive training participants and their 20-PC vectors. **KMeans** minimizes squared distances to cluster centers. Try k = 2, 3, and 4, with 20 initializations and seed 87. Choose the largest mean training silhouette.

What selected two groups?

For a participant, silhouette is (b − a) / max(a, b), where a is mean distance to its own group and b is the smallest mean distance to another group. Mean silhouette is approximately 0.547, 0.414, and 0.202 for k = 2, 3, and 4; k = 2 is retained.

This search never considers k = 1 and has no null test for the existence of discrete populations. It can partition continuous or cohort-driven structure.

What the bootstrap actually repeats

Draw 100 resamples of 73 training TB-positive participants with replacement, using seed 87. For each fixed k, refit KMeans on each resample, assign all original 73 participants using its new centers, and compare assignments with the original fixed-k fit.

Agreement uses the **adjusted Rand index (ARI)**: 1 means identical grouping irrespective of group names, and approximately 0 means chance-level agreement. For k = 2 the median is 1.000, with 10th–90th percentiles about 0.945–1.000.

This resamples clustering only. Gene filtering, scaling, PCA, and k selection are not repeated inside the resamples. These percentiles summarize conditional stability; they are not confidence limits for a biological subtype, and this is not out-of-bag validation.

Assign held-out TB cases using frozen centers

The original two centers assign the 34 test and 35 validation TB-positive participants to their nearest center in all 20 PCs. Figure 10 displays only PC1/PC2 and compares the existing myeloid fractions by assigned group.

Training group 1 contains 39 GHL and 1 R2D2 participant; group 2 contains 33 R2D2 participants. ARI with cohort is about 0.945. This is stable cohort/composition-related structure. Assignment to both centers in validation does not by itself validate distinct disease subtypes.

How cohort and clinical covariates were considered

Figures 11–12 extend the continuous-score regressions. They reuse the same three-gene score and existing plasma fractions, while checking demographic and source-group assumptions.

Figure 11 · Three nested TB models

The outcome remains S, the unrescaled mean of SPI1, MNDA, and LYZ training z scores. Fit these models separately in each split, always with an intercept:

1. S ~ TB + cohort.
2. Add the three broad composition ALRs.
3. Add age/10, male, and HIV-positive indicators.

The third model is the prespecified primary clinical association. Age enters linearly in decades; female and HIV-negative are the reference categories. These fields and exonic reads are complete for all 251 participants, so the nested TB models use the same people.

Primary models have nine parameters in training/test and eight in validation, where cohort is constant. Their residual degrees of freedom are 121, 52, and 52. The figure plots β_{TB} and individual HC3/t 95% intervals. The right panel fits the same adjusted association within each cohort.

Do cohort-specific TB differences agree?

For training and test, a supplemental model interacts cohort with TB and with **every** nuisance slope: the three ALRs, age, male, and HIV status. Its cohort × TB term estimates the adjusted R2D2 TB difference minus the GHL difference.

Allowing all nuisance slopes to differ makes that contrast agree with the two separate cohort fits. A wide or nonsignificant interval does not establish equal associations. No cohort interaction is estimable within the single-cohort validation set.

Technical-depth sensitivity

A separate sensitivity adds $\log_{10}$(exonic reads), centered within the same cohort and split, to the primary model. This is a different metadata field from the assigned-read overlay in PCA/UMAP. It checks dependence on within-cohort depth; it is not a complete technical-artifact correction.

Figure 12 · Separate myeloid components

Replace total myeloid with monocyte, neutrophil, and other myeloid (the remaining five myeloid labels). Keep lymphoid, erythroid/platelet, and the original 27-label Other group. Use five natural-log ratios to that unchanged Other denominator.

Fit S ~ TB + cohort + five ALRs + age/10 + male + HIV-positive. The model has 11 parameters in training/test and 10 in validation. These six grouped parts are nonoverlapping and positive; no pseudocount is needed.

Figure 12 plots the **three myeloid-source coefficients**, not the TB coefficient. Each describes the score difference per one natural-log unit of that source/Other ratio, conditional on the remaining terms. A ratio doubling corresponds to ln(2) times its coefficient.

The TB coefficient from this same model is exported separately. It is approximately 0.170, 0.536, and 0.176 across splits; training and validation intervals cross zero. This sensitivity makes a simple composition-independent TB interpretation less secure.

This grouping differs from Figure 6’s finer model, which still pooled myeloid sources and instead separated erythroid, platelet, and endothelial contributions.

Which tests receive multiplicity correction?

Only the three full-split, fully adjusted primary TB P values form the primary TB family for Benjamini–Hochberg (BH) correction. The within-cohort fits, cohort interactions, source coefficients, and technical-depth sensitivities retain nominal exploratory intervals and P values.

BH controls the expected proportion of false discoveries among declared findings within a defined family, under its assumptions. It does not jointly correct the entire report, earlier gene selection, or prior inspection of these data. HC3 intervals also do not include uncertainty in the estimated source fractions.

How the Xpert-burden analyses were made

Figures 13–14 ask whether the expression signal tracks recorded sputum microbial burden among people already classified as TB-positive. They do not predict TB diagnosis.

Define the clinical measurement

The metadata field semiquant records Xpert Ultra semiquantitative sputum categories: Low, Medium, or High. This is an ordered microbial-burden proxy. It is not blood bacterial load, a direct cfRNA measurement, or a general clinical-severity score.

Restrict to the 142 TB-positive participants. Seven have no recorded category (five training and two test, all GHL), leaving **68 training, 32 test, and 35 validation** participants. Missing values are excluded; neither a missing result nor TB-negative status is coded as zero burden.

Low/Medium/High counts are 16/28/24 in training, 8/8/16 in test, and 8/14/13 in validation. The available-case analysis assumes its associations are informative despite possible selection from missingness.

Figure 13 · Describe the distributions

For each split and burden category, show the unchanged three-gene score. Boxes mark the median and quartiles; whiskers use the usual 1.5-interquartile-range rule. Every participant is also drawn as a point with slight horizontal jitter (seed 87).

Separate boxplot outlier symbols are suppressed to avoid drawing the same observations twice; no extreme participant is removed. Cohort is encoded by marker shape. All panels use the same vertical scale.

The displayed Spearman correlations compare category ranks with score ranks. They are raw monotonic associations: approximately 0.218, −0.105, and 0.098 across splits. They are not the covariate-adjusted slopes in the next figure.

Figure 14 · Estimate a conditional trend

Encode Q = 0, 1, 2 for Low, Medium, High. Regress S on Q + cohort; then add age/10, male, and HIV-positive; then add the three broad ALRs for the primary model. TB is omitted because every participant in this analysis is positive.

$$\mathbf{S_i = \beta_0 + \beta_Q Q_i + \text{covariate terms} + \varepsilon_i}.$$

β_Q is the mean score difference per one category step at fixed modeled covariates. The encoding assumes equal steps in the predictor for Low→Medium and Medium→High; it does not assert equal increments in actual bacterial load.

All three adjustment models within a split use the same available participants. Primary fits have residual degrees of freedom 59, 23, and 27. Points and bars use HC3/t estimates and individual 95% intervals.

Correct only the stated primary family

The three fully adjusted burden P values receive BH correction as a family separate from the three primary TB tests. For sorted P values $p_{(1)} \le \ldots \le p_{(m)}$, the adjusted value at rank i is:

$$\mathbf{q_{(i)} = \min\{1, \min_{j \ge i}[m\, p_{(j)} / j]\}},$$

with m = 3 here. Simpler models and raw correlations remain exploratory. The interval itself is not FDR-adjusted, so a nominal interval can exclude zero while its corrected test does not pass q < 0.05.

Match interpretation to precision

Primary slopes are about 0.201, 0.156, and 0.072 score units per step, with q ≈ 0.137, 0.414, and 0.572. None passes the primary correction. The data do not establish a reproducible adjusted burden trend; limited precision also prevents a claim that no relationship exists.

How genes were ranked for pathway analysis

Figures 15–16 ask whether a predefined biological gene set concentrates near the TB-associated end of a ranked expression list. The genes are not restricted to the classifier's six inputs.

1 · Build an outcome-independent gene universe

Use the training-reference VST and detection filter described on page 37: at least 10 reads in at least 13 training participants. All 11,279 eligible Ensembl IDs enter the pathway analysis; the 1,000-variable-gene restriction applies only to the broad embeddings.

Read gene IDs and names from deseq_results.tsv for annotation only. Its association statistics do not select these genes. When IDs share a symbol, average their VST expression within each participant, yielding **11,264 unique symbols**. No eligible row lacks a symbol.

2 · Define the gene-level regression

For each gene g and each split, fit VST expression on TB status + cohort. Fit a second version adding the three broad composition ALRs from the existing plasma estimates:

$$x_{ig} = \beta_{0g} + \beta_{TB,g}TB_i + \beta_{Cg}C_i + \gamma_{Mg}L_{M,i} + \gamma_{Lg}L_{L,i} + \gamma_{EP,g}L_{EP,i} + \varepsilon_{ig}.$$

The cohort-only model omits the three γ terms. Validation has only TBSQUARE, so it omits C. These models do not include age, sex, HIV, or read depth. Their adjustment scope differs from the clinical primary model.

3 · Rank by association relative to uncertainty

$$r_g = \beta_{TB,g} / SE_{HC3}(\beta_{TB,g}).$$

Sort genes from largest to smallest r. Positive values indicate higher adjusted expression in TB-positive participants; negative values indicate lower expression. This is an OLS/HC3 statistic, not a DESeq2 Wald statistic. No gene-level P-value cutoff is applied before enrichment.

There are 67,584 gene fits (11,264 × 3 splits × 2 adjustments), producing six ranked lists. Current exported rankings contain no ties.

4 · Specify the biological gene sets [M7]

The pinned human MSigDB 2026.1 definitions include all 50 Hallmark sets plus Reactome neutrophil degranulation. A tested set must have 15–500 members in the ranked universe. Pancreatic beta cells has only 12 eligible members and is excluded.

Thus each enrichment run tests **50 sets: 49 Hallmarks and one Reactome set**. The five focused themes are interferon-alpha response, interferon-gamma response, TNF signaling via NF-κB, inflammatory response, and neutrophil degranulation.

5 · Repeat after removing the six model genes

Drop GBP5, AHNAK, JMJD1C, MNDA, LYZ, and SPI1 from the entire ranked universe, reducing it to 11,258 symbols. Recalculate enrichment using that list. Other genes' regression fits do not change, so they do not need refitting.

Among the focused sets, only neutrophil degranulation loses members: MNDA and LYZ, reducing its eligible size from 404 to 402. The other focused sizes remain 91, 185, 157, and 132. Removing genes elsewhere in the ranking can still alter enrichment for a set that loses no members.

This exclusion checks direct dependence on the classifier genes. It does not remove correlated genes, change the deconvolution fractions, or create an independent biological replication. The alpha and gamma sets themselves share 69 eligible genes.

Which runs appear in the figures?

Figure 15 shows the five focused sets under both adjustments and both gene universes (four panels). **Figure 16** shows 16 Hallmarks: the eight lowest and eight highest training cohort-only NES values. Its two panels use all eligible genes, comparing the two adjustments.

The broader overview therefore has training-selected display rows. All 50 eligible sets remain in the numerical results and in each run's FDR calculation.

How GSEA, NES, and its FDR were calculated

Preranked gene-set enrichment analysis (GSEA) evaluates the positions of a set's members in a fixed ranked gene list. Its null model differs from participant-level regression. [M7]

1 · Walk down the ranked list

For a set S with m members among N ranked genes, begin a running sum at zero. At a member gene, step up by $|r_g|$ divided by the sum of |r| over all members. At a nonmember, step down by 1/(N−m). The executed weighting exponent is 1.

**$R_S(k) = \Sigma_{j \le k,\, g_j \in S} |r_j| / \Sigma_{g \in S} |r_g|$
− count(nonmembers among first k) / (N − m)**.

The **enrichment score (ES)** is the signed excursion with the greatest absolute magnitude. A positive ES means members concentrate near the positive-TB end; a negative ES means concentration near the negative end. Every member need not change in the same direction.

2 · Generate a gene-label reference

GSEApy 1.1.13 uses 1,000 permutations with seed 87. Gene identities are shuffled against the fixed ranked-statistic positions. Set sizes and the distribution of rank values remain fixed; the same shuffled ordering is shared across sets within a run.

Participants and TB labels are not permuted. The procedure does not refit regressions, resample expression vectors, or propagate uncertainty in the ranks. It asks whether these members occupy unusually extreme positions compared with random gene assignments.

3 · Normalize the enrichment score

**$NES_S = ES_S$ / mean(|null ES_S|
among permutations of the same sign)**.

Positive and negative null scores are handled separately. NES is not a z score: the null mean is not subtracted, and the denominator is not a standard deviation. It is also not an expression fold change, patient score, or probability.

4 · Estimate empirical GSEA FDR

The nominal permutation P value is the proportion of same-sign null scores at least as extreme as the observed ES. The reported "FDR q-val" uses a different calculation: pool normalized null scores across the 50 tested sets in that run.

For positive observed NES z, the estimate is the following ratio, capped at one:

**[fraction of nonnegative null NES ≥ z]
÷ [fraction of nonnegative observed NES ≥ z]**.

Negative scores use the analogous negative-side lower tail. This is **empirical GSEA FDR, not BH adjustment** of nominal P values. The installed native implementation does not apply its optional monotonic adjustment. An exported zero means no sufficiently extreme tail was observed in this finite calculation, not zero error probability.

What is the correction family?

There are 12 runs: 3 splits × 2 adjustment sets × 2 gene universes, producing 600 pathway rows. Each run calculates its own FDR over 50 sets. There is no further correction across all 12 runs.

Figures 15–16 mark cells with within-run GSEA FDR < 0.05. Focused panels still use the 50-set family, not only the five displayed pathways. Color represents NES on a common −3.5 to +3.5 scale.

What the permutation model leaves open

Gene-label permutations do not preserve participant-level expression covariance through resampling. Strongly correlated pathway genes can therefore yield more optimistic enrichment evidence than a participant-based sampling test. The fixed ranked list and overlapping sets also mean the 50,000 null NES values are not independent experiments.

A significant NES supports unusual relative concentration in this ranking. It does not establish a causal pathway mechanism or precisely quantify an average expression difference between TB groups. The next analysis addresses that different quantity.

Sources: gene_interactions_pathways.ipynb · gp.prerank; GSEApy 1.1.13 native statistics/algorithm source [M7]

How the participant-level pathway scores were made

Figure 17 summarizes pathway expression in each person, then estimates its adjusted TB-associated difference. This complements enrichment by retaining the participant as the unit of statistical replication.

1 · Define the score genes

For each of the five focused themes, use every eligible member after excluding the six XGBoost genes. Do not select only the members driving the greatest GSEA excursion. The resulting sizes are 91, 185, 157, 132, and 402 genes.

2 · Standardize each gene using training

$$z_{ig} = (x_{ig} - \text{mean}_{train} x_g) / \text{SD}_{train}(x_g).$$

Here x is the training-reference VST. This notebook uses sample SDs (ddof = 1, denominator 129 for 130 training participants). The same training quantities transform test and validation.

3 · Average, then rescale the composite

$$u_{iS} = \text{mean}_{g \in S}(z_{ig})$$
$$P_{iS} = u_{iS} / \text{SD}_{train}(u_S).$$

All genes have equal positive weights. The mean of u in training is already zero, so no second explicit centering is needed. The final P has training mean zero and sample SD one. A difference of 0.8 is 0.8 training SDs of this pathway composite.

This differs from the three-gene score, which was not rescaled after averaging. It is also not SHAP, ssGSEA, GSVA, or an optimized predictor. It is a fixed aggregate expression summary, not a direct measurement of pathway activation.

4 · Fit the same two adjustment designs

Regress P on TB + cohort, then add the three broad composition ALRs, separately in each split. Age, sex, and HIV are not added. β_{TB} estimates the adjusted positive-minus-negative difference in training-composite-SD units.

5 · Show uncertainty and correct 30 tests

The dots are OLS TB coefficients; bars are individual HC3/t 95% intervals using residual degrees of freedom. Five pathways × three splits × two adjustments yield **30 participant-score tests**. BH is applied to all 30 two-sided P values together.

A star in Figure 17 means BH-adjusted q < 0.05 in that 30-test family. This differs from GSEA's empirical FDR. The intervals remain pointwise rather than simultaneous or FDR-adjusted, so an interval can exclude zero without a star.

Why the results can differ from GSEA

GSEA asks about the relative positions of a pathway's genes among all ranked genes. The score regression asks how much the average pathway summary differs between participants with different TB status at the same covariate values.

For validation neutrophil degranulation after broad composition adjustment, leave-six-out GSEA has NES ≈ 1.864 and empirical FDR ≈ 0.00245. Yet the participant-score difference is approximately **0.364 training SD**, with 95% CI **−0.195 to 0.923** and BH q ≈ 0.212.

These are compatible: a set can occupy the positive end of the gene ranking while its average score varies enough between people to leave the adjusted mean difference imprecise. The two statistics are not competing estimates of the same parameter.

What a replicated score association supports

A consistently positive adjusted interferon-gamma score across splits supports a broader expression program extending beyond the six classifier genes. It remains a conditional association with estimated composition, not proof of intracellular signaling in a particular cell type.

Genes are correlated components of each person's score, not independent participant replicates. Regression uncertainty is conditional on the fixed training transform, gene-set definition, and estimated source fractions; it does not propagate uncertainty in those upstream choices.

How the optional gene-exclusion deconvolution works

The primary figures already use the existing manuscript plasma estimates. This separately configured sensitivity analysis asks whether the six selected genes directly influence reconstructed RNA-source estimates.

The mixture being modeled [M8]

Deconvolution relates each observed mixture of transcript counts to reference expression profiles from labeled cells. Conceptually, a sample's gene probability is a weighted mixture:

$$\Pr(\text{gene } g \mid \text{sample } i) = \Sigma_t \, \theta_{it} \, \psi_{tg},$$

where θ_{it} is the relative RNA contribution of source t and ψ_{tg} its reference gene-expression probability. BayesPrism uses count data, prior/reference information, and posterior sampling to estimate such contributions and update reference profiles.

A posterior is the model's distribution of plausible values after conditioning on its inputs. These are RNA contributions, not absolute cell counts. Reference mismatch and differences in RNA release can still affect interpretation.

Prepare and align the reference

The recovered reference contains 9,834 cells grouped into 41 source labels. After the configured cleanup, 29,546 genes remain; 29,435 overlap the mixture counts. Trailing Ensembl version suffixes are removed from both sides, with uniqueness and target-gene checks.

Reference cell counts are **summed within each source label** to make aggregated gene-expression profiles. The cleanup uses the package's configured ribosomal, mitochondrial, sex-chromosome and MALAT1 exclusions and requires positive counts in at least five reference cells before aggregation (exp.cells = 5). The resulting profiles enter BayesPrism as GEP input.

Create a matched pair for every participant

First fit the participant using the recovered reference and available raw counts. Then repeat after removing GBP5, AHNAK, JMJD1C, MNDA, LYZ, and SPI1. Both fits begin with the same reference, settings, and seed; their updated references may subsequently differ.

Keep the other filtering rules matched

The baseline sample-specific outlier filter uses outlier.cut = 0.01 and outlier.fraction = 0.1. Its retained gene set is frozen for both variants; no second differential outlier screen is introduced. Assertions require all six genes at baseline and exactly six fewer genes in the excluded fit.

This controls an avoidable difference between the two fits. It does not make the reconstructed pipeline an exact historical reproduction, because the original filtering branch and software history remain incompletely recovered.

Configured inference and resources

BayesPrism 2.2.3 runs initial Gibbs sampling, a MAP reference update, then final Gibbs sampling. Each sampling stage has 1,000 iterations, 500 burn-in iterations, thinning by two, seed 123, and α = 1. This retains 250 draws per stage; draws are not independent participants.

Gibbs sampling repeatedly samples conditional model components. Burn-in discards the early part of the chain; thinning retains every other subsequent draw. MAP selects the most probable updated reference under its model. These settings describe the computation, not a demonstration of convergence.

The approved run uses up to eight sample workers with one numerical thread each and a 32 GiB memory cap. Completed pairs are checkpointed. Final type-level fractions are taken from the final sampling stage.

How the completed comparison will be read

Compare baseline versus excluded fractions within the same participant, then repeat the raw/partial rank associations and broad-composition TB regression. Small changes would reduce concern about direct use of these six markers; large changes would identify sensitivity to their inclusion.

Neither outcome establishes independence from correlated genes or validates absolute source abundance. The report's original pending-status statement is retained: no unfinished fit is treated as a result, and the primary analysis does not depend on this optional check.

What was checked, and what the checks mean

This edition adds methodological detail without changing the original figures, interpretations, or fitted analyses. The checks below distinguish numerical consistency from biological evidence.

Analysis layer	Checked against saved implementation and outputs	What remains conditional
Expression and participants	Sample-ID alignment; separate normalization paths; training scaling; exact VST-to-SHAP expression agreement.	Normalization and RNA-source reference adequacy; batch-dependent preprocessing in the original GFS analysis.
SHAP accounting	1,506 gene rows = 251 × 6. Probability reconstruction error < 1.86×10^{-7}; interaction row-sum error < 3.52×10^{-7}.	The fitted model, selected genes, and tree-path reference rule. Additivity does not establish causality or calibration.
Expression patterns	Training feature/scaler/PCA fits; held-out transformations; all-20-PC clustering; conditional bootstrap scope.	Cohort/technical structure and upstream representation choices. Stable groups are not automatically biological subtypes.
Clinical and source regressions	Outcome units; composition denominators; demographic coding; HC3/t intervals; available-case burden analysis; BH families.	Model form, working independence, measured covariates, and uncertain same-assay deconvolution estimates.
Pathways	Gene universe and mapping; six-gene exclusions; 12 GSEA runs; empirical FDR versus BH for 30 score tests.	Curated definitions, gene dependence, rank uncertainty, and fixed transforms. NES and score differences answer distinct questions.

Where the calculations can be inspected

The source notebooks are in Scripts_For_Manuscript: **gene_interactions.ipynb** (model and preprocessing); **gene_interactions_interpretation.ipynb** (SHAP and initial sources); **gene_interactions_patterns.ipynb**; **gene_interactions_cohort_clinical.ipynb**; **gene_interactions_pathways.ipynb**; and the optional **gene_interactions_deconvolution.ipynb**.

Saved figures and tables are in Figure_Output/Gene_Interactions/focused. The interpretation, patterns, clinical, and pathway provenance JSON files record input hashes and package versions. The methods in this companion describe those local executed versions.

A distinction that applies throughout

The test and validation labels are withheld from the programmed XGBoost feature search and fit. They are then used for model evaluation and for the biological association regressions shown in this report. Those are different uses of the data.

Because these datasets and results have already been inspected, the follow-ups are exploratory. Separate panels preserve the splits and reveal consistency or heterogeneity; they do not reset the data to a prospectively untouched confirmatory experiment.

Numerical checks support accurate description of what was calculated. Mechanistic claims still require independent biological evidence, while new predictive claims require appropriately protected evaluation data.

Methods references and implementation notes

The original report's sources remain on page 27. These additional references support the methodological explanations; local code and saved outputs determine the exact settings used here.

M1 · SHAP's original formulation

Lundberg and Lee (2017), *A Unified Approach to Interpreting Model Predictions*. Introduces SHAP and its Shapley-value attribution framework. Read the primary paper.

M2 · TreeSHAP and interactions

Lundberg, Erion, and Lee (2018), *Consistent Individualized Feature Attribution for Tree Ensembles*. Tree attribution and interaction methods. Lundberg et al., *From local explanations to global understanding with explainable AI for trees*. Primary manuscript.

M3 · XGBoost's exact local implementation

Prediction, contribution, and interaction API; tree parameters; initial intercept. Version 3.0.5 source: tree-path algorithm, baseline and pair construction, Hessian cover, and logistic curvature.

M4 · Count normalization and VST

Love, Huber, and Anders (2014), *Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2*. Primary paper; official vignette. The current vignette is newer than the local R 4.0.5 / DESeq2 1.30.1 export; local saved coefficients and code govern the report's calculations.

M5 · Robust regression uncertainty

statsmodels 0.14.6 HC3 covariance documentation. The notebooks explicitly set use_t=True and use residual degrees of freedom. Multiple-testing families are defined in the local clinical and pathway code; they should not be inferred from a plot alone.

M6 · Embedding and clustering

McInnes, Healy, and Melville, *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. Primary method; transforming new data; clustering caveats. Local versions are umap-learn 0.5.7 and scikit-learn 1.6.1. KMeans, silhouette, and adjusted Rand index.

M7 · Enrichment and gene-set definitions

Subramanian et al. (2005), *Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles*. Primary paper; official guide. Pinned MSigDB 2026.1 Hallmark and Reactome definitions. GSEApy 1.1.13 native NES/P-value/FDR calculations and running-sum/permutation implementation.

M8 · Reference-based deconvolution

Chu et al. (2022), *Cell type and state inference from bulk and single-cell RNA-seq*. BayesPrism primary paper; official repository. The optional reconstruction uses installed version 2.2.3. Its settings describe a sensitivity analysis; they do not establish the precise historical settings of the existing plasma table.